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Method Article

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Posted Date: June 4th, 2026

DOI: https://doi.org/10.21203/rs.3.rs-9903418/v1

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Additional Declarations: The authors declare no competing interests.

A Zero-Trust Agentic AI Methodology for Cyber-Resilient Energy Market Clearing under Residual Cyber Contamination and Dynamic Uncertainty

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Abstract

Real-time energy market clearing increasingly depends on cyber-physical data streams, including bids, measurements, dispatch acknowledgments, flexibility responses, and settlement records. When these inputs are affected by cyber contamination, degraded observability, or dynamic uncertainty, conventional clearing models may admit unreliable data, generate infeasible dispatch outcomes, distort price signals, or weaken the evidentiary basis of settlement. This paper proposes a zero-trust agentic AI methodology for cyber-resilient energy market clearing under residual cyber contamination and dynamic uncertainty. The proposed methodology introduces a bounded agentic verification layer in which specialized agents evaluate market-critical data streams, estimate trust, quantify residual cyber contamination, assess uncertainty pressure, and determine admissibility before inputs are passed to the market-clearing optimizer. The framework integrates calibrated agentic consensus, channel-level admissibility screening, feasible-domain contraction, cyber-risk-aware objective terms, settlement-legitimacy modeling, audit-record construction, and human-governed escalation and return-to-normal certification. A controlled benchmark validation based on modified IEEE test systems is used to evaluate the internal consistency and operational behavior of the proposed methodology under cyber-contaminated and uncertainty-intensive scenarios. The results indicate that the proposed approach reduces residual cyber-risk exposure and false admission of contaminated inputs while improving feasibility preservation, settlement legitimacy, and auditability compared with conventional, cyber-aware, and rule-based zero-trust clearing strategies. The framework also maintains computational practicality by keeping the additional verification and auditability overhead modest relative to the network-constrained optimization process. The proposed methodology provides a structured foundation for auditable, human-governed, and cyber-resilient market operation in renewable-rich and flexibility-intensive energy systems.

Keywords: Zero-trust security, agentic AI, multi-agent systems, cyber-resilient energy markets, real-time market clearing, smart grids, residual cyber contamination, admissibility screening, settlement legitimacy, critical infrastructure protection.

1. Introduction

Real-time energy market clearing is increasingly exposed to cyber-physical uncertainty because modern power systems depend on digitally mediated bids, offers, measurements, dispatch acknowledgments, flexibility responses, and settlement records. In renewable-rich and flexibility-intensive systems, these information streams are not only operational inputs but also market-critical evidence used to determine dispatch, prices, reserve activation, demand response continuation, and settlement legitimacy[1]. This creates a central research problem: conventional market-clearing formulations generally assume that input data are sufficiently reliable,

whereas cyber-contaminated data may distort market prices, trigger infeasible dispatch, manipulate flexibility activation, or weaken the evidentiary basis of settlement[2]. Recent work on trust-aware federated deep reinforcement learning has shown that cyber-resilience in flexible microgrid markets requires explicit trust modeling rather than purely economic optimization [3, 4]. Similarly, secure pricing and trust-based demand response models demonstrate that market-clearing decisions must account for the credibility of flexibility resources and real-time cyber conditions [5]. Building on this direction, this study proposes a zero-trust agentic AI framework for cyber-resilient energy market clearing, in which autonomous verification agents evaluate trust, residual cyber contamination, uncertainty pressure, admissibility, settlement legitimacy, and return-to-normal readiness before market data are admitted into the clearing optimizer [6].

The proposed framework is motivated by the growing convergence of agentic AI [7], blockchain-governed trust infrastructures, federated intelligence, and cyber-resilient energy market design. Blockchain-based architectures have been explored as mechanisms for improving transparency, sustainability governance, and decentralized coordination, including SDG-oriented digital platforms, NFT-enabled loyalty mechanisms in agribusiness , and conceptual Ethereum-based models for sustainable development implementation [8]. Although these studies are not energy-market-clearing models, they show the relevance of auditable, decentralized, and trust-sensitive digital infrastructures for governance-oriented applications. In the energy domain, blockchain-governed [9] federated intelligence has been proposed as a formal model for auditable collaborative learning, indicating that decentralized trust and traceability can support secure AI coordination across distributed participants . Risk-aware multi-agent reinforcement learning and blockchain enforcement have also been studied for peer-to-peer trading and demand response in residential energy communities, showing the importance of combining autonomous decision-making with enforceable market rules [10, 11]. These foundations motivate the present study's use of zero-trust verification and auditability as embedded components of market clearing rather than external cybersecurity add-ons.

A second motivation arises from recent advances in LLM-assisted and multi-agent energy-system intelligence [12, 13]. Retail-informed secure wholesale market models have shown that LLM-assisted multi-agent deep reinforcement learning can support bid generation and market coordination in networked microgrids [14]. Safe, secure, and uncertainty-aware demand response architectures further suggest that LLM-guided hierarchical multi-agent systems can coordinate energy resources under cyberattacks and uncertainty when constrained by operational safety rules [15, 16]. In parallel, risk-aware secure microgrid pricing models validated with hybrid LLM-blockchain structures demonstrate that LLM-supported decision processes can be combined with cyberattack-aware pricing and hybrid storage flexibility [17]. Beyond energy systems, real-time trust-weighted consensus and auditing of LLMs for secure propaganda detection show that LLM outputs can be governed through reputation modeling, explainability, and consensus-based auditing [18]. These studies support the central assumption of this paper: agentic AI can be useful in cyber-resilient energy markets only when its actions are bounded, auditable, explainable, and coupled to formal decision rules.

The proposed model also builds on a broader methodological background in optimization, learning, and cyber-physical resilience. Optimization-oriented work on TSP-based nested clustering for multi-depot heterogeneous fleet routing illustrates the value of structured decomposition and scalable decision modeling in complex operational systems [19] . Deep neural network applications for fatigue lifetime estimation show the usefulness of data-driven prediction in engineering systems where degradation, uncertainty, and reliability are central concerns . In intelligent transportation systems, hybrid LLM-based emergency management with constraint-aware edge deployment demonstrates how LLM-based reasoning can be constrained for time-sensitive infrastructure applications . These studies collectively reinforce the need for models that com-

bine optimization, learning, constrained autonomy, and reliability-aware decision support. In the present paper, this logic is transferred to real-time energy market clearing by designing a framework in which agentic AI performs verification and recommendation, while the final market outcome remains governed by mathematical constraints, feasible-domain contraction, and human approval rules.

Within energy-market resilience, several recent studies provide the immediate conceptual foundation for the proposed framework[20]. Post-disruption resilient recovery of microgrid real-time balancing markets has emphasized the importance of market-state degradation, recovery-oriented operation, and dynamic uncertainty in post-contingency clearing [21]. Resilience-oriented market-to-delivery demand response research has shown that constrained deep reinforcement learning can support flexible energy delivery under cyber-physical disruptions and uncertainty [22]. Energy storage-based demand response models have further highlighted the combined effect of time-varying uncertainty, cyber-degraded observability, and settlement risk in renewable-rich integrated energy systems [23, 24]. Credibility-aware demand response continuation and regime-switching market operation has shown that market continuation should depend on cyber credibility, dynamic uncertainty, and hybrid threat severity rather than a binary normal/failure view [25, 26]. Heterogeneous energy storage-supported demand response studies also indicate that compound disturbances require resilient market-to-delivery frameworks rather than isolated dispatch correction mechanisms [27]. The present study extends these ideas by introducing a zero-trust agentic AI layer that explicitly determines whether market inputs should be admitted, restricted, escalated, or excluded before clearing.

The main contribution of this paper is a compact but integrated formulation for zero-trust agentic AI-based market clearing. First, the paper develops a multi-agent verification layer for bids, measurements, dispatch acknowledgments, flexibility responses, and settlement logs. Second, it formulates residual cyber contamination, dynamic uncertainty pressure, and admissibility confidence as coupled but distinct constructs. Third, it embeds these indicators into a contracted feasible domain and cyber-resilient clearing objective, allowing market continuation under degraded trust while avoiding unsafe normal clearing. Fourth, it introduces settlement legitimacy and audit-record construction as formal components of the clearing process, ensuring that cyber-resilient dispatch decisions remain financially and operationally defensible. Fifth, it defines human-governed escalation and return-to-normal certification rules so that agentic AI supports critical infrastructure decision-making without becoming an unconstrained autonomous operator[28].

Representative simulation-based results indicate that the proposed framework can reduce residual cyber-risk exposure, lower false admission of contaminated inputs, improve settlement legitimacy, and preserve feasible market operation under cyber-degraded conditions[29]. Compared with conventional real-time clearing, the proposed model introduces moderate operating cost and computational overhead but improves cyber resilience, admissibility control, and auditability. Compared with rigid rule-based zero-trust screening, it avoids excessive conservatism by using calibrated trust, agentic consensus, residual contamination, and uncertainty pressure rather than fixed rejection rules. Nevertheless, the study has several limitations. The evaluation is based on benchmark systems and representative cyberattack scenarios rather than real utility cyber-physical logs. The reliability of the agentic verification layer depends on the quality of evidence streams, historical baselines, and audit data. The current formulation also assumes bounded adversarial capability and does not yet address fully compromised multi-agent verification or attacks directly targeting the agentic AI layer itself. These limitations define clear directions for future work, including validation on larger test systems, digital-twin-based cyber-physical environments, adversarial testing of LLM-guided agents, and integration with real market operator workflows[30].

The remainder of this paper is organized as follows. Section II reviews the state of the art

in cyber-resilient energy markets, zero-trust architectures, agentic AI, federated intelligence, blockchain-based auditability, and uncertainty-aware market operation. Section III presents the proposed methodological framework, including the zero-trust agentic verification layer, admissibility model, cyber-resilient clearing formulation, governance constraints, and auditability model . Section IV reports the method validation and comparative evaluation, and Section V discusses the implications, limitations, future research directions, and conclusion.

2. State of the Art

2.1. Cyber-Resilient Energy Markets and Trust-Aware Market Operation

The increasing digitalization of energy markets has transformed real-time clearing from a purely economic optimization problem into a cyber-physical decision process. Market operators now depend on bids, offers, measurements, dispatch acknowledgments, flexibility responses, and settlement records that may be exposed to cyber manipulation or degraded observability. This creates a direct connection between cybersecurity, market feasibility, price formation, and settlement legitimacy. Early cyber-physical power-system security studies showed that false data injection attacks can manipulate state estimation while bypassing conventional bad-data detection, making cyber-contaminated measurements a serious threat to operational decision-making [4]. In market environments, similar vulnerabilities may affect not only state estimation but also balancing actions, demand response activation, redispatch, and financial settlement.

Recent studies have begun to address this problem by incorporating trust, cyber resilience, and uncertainty into energy market models. Sepehrzad et al. developed a trust-aware federated deep reinforcement learning framework for flexible energy markets in microgrids, showing that cyber-resilient market operation requires explicit modeling of trust rather than relying only on economic optimization [31, 3]. In a related contribution, secure microgrid pricing was integrated with trust-based demand response through a multi-layer federated deep reinforcement learning architecture, emphasizing that demand-side flexibility should be treated as a credibility-sensitive resource under cyber uncertainty [32, 5]. These works establish an important foundation for the present study, but they do not fully formalize a zero-trust admission layer that screens every market-critical input before clearing.

A further stream of research has examined recovery-oriented and regime-switching energy market operation. Post-disruption resilient recovery of microgrid real-time balancing markets has shown that market operation after disturbance should not be simplified into a binary normal/failure condition; instead, intermediate degraded states require specific recovery-oriented clearing logic [33, 21]. Credibility-aware demand response continuation and regime-switching market operation has similarly argued that demand response should continue only when cyber credibility and uncertainty conditions remain within defensible limits [26]. Resilience-oriented market-to-delivery demand response frameworks further connect cyber-physical disruptions with constrained reinforcement learning and delivery-side feasibility [22]. These studies motivate the current paper's use of normal, zero-trust cyber-resilient, and fallback regimes.

Energy storage and flexible resources have also become central to cyber-resilient market design. Energy storage-based demand response under time-varying uncertainty, cyber-degraded observability, and risk-aware settlement shows that storage and demand response can preserve system flexibility but may also introduce additional settlement and observability risks [24]. Heterogeneous energy storage-supported demand response extends this line by considering compound disturbances in integrated energy systems [27]. These studies support the need for a clearing model that does not overuse flexibility during cyber-degraded operation and that explicitly penalizes flexibility over-reliance when response credibility is uncertain.

2.2. Zero-Trust Security and Critical Infrastructure Protection

Zero-trust security has emerged as a major cybersecurity paradigm for environments where implicit trust in users, devices, services, or internal network locations is no longer defensible. NIST SP 800-207 defines zero-trust architecture as a cybersecurity approach that shifts protection away from static network perimeters and toward resource-level, continuously evaluated access decisions [1]. This principle is highly relevant to energy markets because market-clearing inputs are analogous to high-impact digital resources: they should not be trusted merely because they originate from an apparently authorized participant, device, platform, or internal communication channel.

However, most zero-trust research has focused on enterprise networks, access control, cloud services, and identity management rather than market-clearing optimization. In critical infrastructure, zero trust must be extended beyond access authorization to include data admissibility, operational feasibility, economic defensibility, and auditability. For energy markets, this means that every bid, measurement, dispatch acknowledgment, flexibility response, and settlement record should be treated as conditionally admissible rather than automatically valid. The present study advances this perspective by translating zero-trust logic into a mathematical market-clearing framework in which trust scores, residual contamination, uncertainty pressure, and human escalation jointly determine whether data can enter the optimizer.

This extension is important because cyber incidents in energy markets may not immediately require full market suspension. In many cases, the system may operate in a degraded but recoverable state where normal clearing is unsafe, but fallback intervention is too conservative. Existing recovery-oriented market studies have already identified the need for intermediate operating regimes under post-disruption uncertainty [21, 26]. The proposed zero-trust framework builds on this idea by defining an admissibility-driven clearing mode between normal operation and fallback intervention.

2.3. Agentic AI, Multi-Agent Systems, and LLM-Assisted Energy Intelligence

Multi-agent systems are increasingly relevant for cyber-resilient energy markets because energy systems involve multiple interacting entities, including market operators, generators, aggregators, storage units, flexible loads, cyber monitors, and settlement authorities. Multi-agent reinforcement learning has been widely studied for settings where agents must coordinate or compete under partial observability and dynamic interactions. The multi-agent actor-critic framework introduced by Lowe et al. is a foundational contribution because it addresses the non-stationarity and coordination difficulties that arise in multi-agent learning environments [34]. This is conceptually relevant to energy markets, where agents must reason over interdependent market, cyber, and physical states.

Recent energy-market studies have begun to incorporate LLM-assisted and multi-agent intelligence. Retail-informed secure wholesale energy market modeling has used LLM-assisted multi-agent deep reinforcement learning for bid generation and projection-based clearing exchange in networked microgrids [14]. Safe, secure, and uncertainty-aware demand response architectures have explored LLM-guided hierarchical multi-agent deep reinforcement learning under uncertainty and cyberattacks [16]. Risk-aware secure microgrid pricing with hybrid LLM-blockchain validation further shows that LLM-assisted reasoning can be connected with cyberattack-aware pricing, demand response flexibility, and hybrid energy storage [17]. These studies show the potential of LLM-guided multi-agent intelligence in energy systems, but they still require stronger governance and admissibility mechanisms to prevent unsafe autonomous intervention.

Outside the energy domain, trust-weighted consensus and auditing of LLMs for secure propaganda detection demonstrates the importance of reputation modeling, explainable AI, and

consensus-based validation when multiple LLMs or AI agents are used in security-sensitive environments [18]. This is directly relevant to the present paper because agentic AI in critical infrastructure should not be treated as a black-box decision maker. Instead, it should generate auditable evidence, trust scores, disagreement indicators, and operator-readable explanations. Therefore, the proposed framework uses agentic AI as a bounded verification and recommendation layer, not as an unconstrained autonomous market operator.

2.4. Federated Intelligence, Blockchain Governance, and Auditability

Distributed energy markets require collaborative intelligence across participants that may not be willing or legally allowed to share raw operational data. Federated learning and blockchain-based governance have therefore become important tools for privacy-preserving and auditable coordination. Blockchain-governed federated intelligence has been proposed as a formal model for trust-aware and auditable collaborative learning, emphasizing the value of decentralized verification, model accountability, and traceable cooperation . In energy markets, these features are particularly valuable because participants may contribute data, bids, forecasts, and flexibility responses while retaining local control over sensitive information.

Blockchain-based systems have also been studied in broader sustainability and governance contexts. A blockchain-based platform for supporting the implementation of the SDGs illustrates how distributed ledgers can support transparency and coordination across institutional actors . A conceptual hybrid model based on Ethereum for implementing sustainable development goals further highlights blockchain's role in transparent governance and decentralized coordination [8]. NFT-based blockchain mechanisms have also been explored for loyalty and incentive design in agribusiness, showing how tokenized and auditable incentives can support behavioral engagement . Although these studies are outside electricity-market clearing, they provide useful conceptual support for traceability, incentive compatibility, and governance-oriented digital infrastructures[35].

In energy communities, risk-aware multi-agent reinforcement learning combined with blockchain enforcement has been proposed for peer-to-peer trading and demand response [36, 11]. This stream demonstrates that decentralized enforcement and autonomous learning can be combined for distributed energy markets. However, blockchain enforcement alone does not solve the real-time admissibility problem: market operators still need to decide which bids, measurements, flexibility signals, and settlement evidence can be trusted during cyber-degraded operation. The proposed zero-trust agentic AI framework addresses this gap by using auditability as a formal component of clearing and settlement legitimacy rather than only as a post-hoc record-keeping mechanism[37].

2.5. Cyber-Physical Uncertainty, Flexibility, and Settlement Risk

Renewable-rich energy systems are affected by multiple forms of uncertainty, including renewable forecast error, demand deviation, price volatility, flexibility non-delivery, and cyber-degraded observability. These uncertainties interact with cyber risk because contaminated measurements or manipulated flexibility signals can become more damaging when the system is already operating near physical or economic limits. Recent studies on demand response and integrated energy systems have shown that uncertainty-aware market operation is essential for resilient performance [38, 24]. Similarly, secure microgrid pricing studies have shown that demand response and flexibility should be priced and activated with trust and security considerations in mind [39, 5].

Settlement risk is another underdeveloped area in cyber-resilient market clearing. A dispatch decision may be technically feasible but financially disputable if bids, measurements, execution records, or audit logs are degraded. This is especially important in real-time balancing markets, where settlement depends on the integrity of high-frequency operational evidence. Recent work

on cyber-degraded observability and risk-aware settlement highlights this issue in renewable-rich integrated energy systems [40, 24]. The present study extends this direction by defining settlement legitimacy as a modeled index that influences the clearing objective and return-to-normal certification.

2.6. Optimization, Simulation, and Benchmarking Foundations

The proposed framework requires a reproducible simulation environment for network-constrained market clearing. MATPOWER is a widely used open-source package for power flow, optimal power flow, and power-system research, making it suitable for testing market-clearing formulations on standard IEEE systems [31]. IEEE 30-bus, IEEE 118-bus, and IEEE 300-bus systems can support staged evaluation, from baseline validation to scalability testing. These benchmark systems allow the proposed model to be evaluated under congestion, redispatch, renewable uncertainty, and cyberattack scenarios in a controlled and reproducible manner.

Constraint-aware LLM deployment in intelligent transportation emergency management further supports the use of bounded AI reasoning in critical infrastructure contexts . This is relevant because the proposed energy-market framework similarly treats LLM-guided or agentic AI as a constrained decision-support mechanism. The model does not allow agents to directly perform high-impact actions such as load shedding, return-to-normal certification, or final settlement authorization; instead, these actions are subject to formal constraints and human approval.

2.7. Research Gap and Positioning of the Present Study

The reviewed literature shows clear progress in cyber-resilient energy markets, trust-aware demand response, federated intelligence, blockchain auditability, LLM-assisted multi-agent learning, and uncertainty-aware market operation[41]. However, four gaps remain unresolved. First, existing cyber-aware market models often treat cybersecurity as a detection or penalty layer rather than as a formal data-admissibility mechanism before clearing. Second, current LLM-assisted and multi-agent energy models provide promising decision-support capabilities but do not sufficiently formalize safe agent action sets, human escalation, and return-to-normal certification. Third, blockchain and federated intelligence studies improve auditability and collaboration, but they do not directly determine how cyber-contaminated market inputs should be admitted, restricted, or rejected during real-time clearing[42]. Fourth, settlement legitimacy is still rarely modeled as an explicit objective or constraint in cyber-resilient market-clearing formulations.

This paper addresses these gaps by proposing a zero-trust agentic AI framework for cyber-resilient energy market clearing. The framework integrates multi-agent verification, trust calibration, residual cyber-contamination modeling, uncertainty-pressure assessment, admissibility screening, feasible-domain contraction, settlement-legitimacy modeling, audit-record construction, and human-governed regime transition. In this way, it advances the state of the art from cyber-aware optimization toward governed, auditable, and zero-trust market operation under degraded digital trust.

3. Methodological Modeling Framework

3.1. System Overview

The proposed framework models real-time energy market clearing as a zero-trust cyber-physical decision process in which market inputs are not directly admitted into the clearing engine. Instead, each bid, measurement, dispatch signal, flexibility response, and settlement record is first evaluated by a coordinated agentic AI verification layer. The purpose is to prevent cyber-contaminated information from entering the market-clearing problem while preserving operational feasibility under renewable, load, price, and flexibility uncertainty.

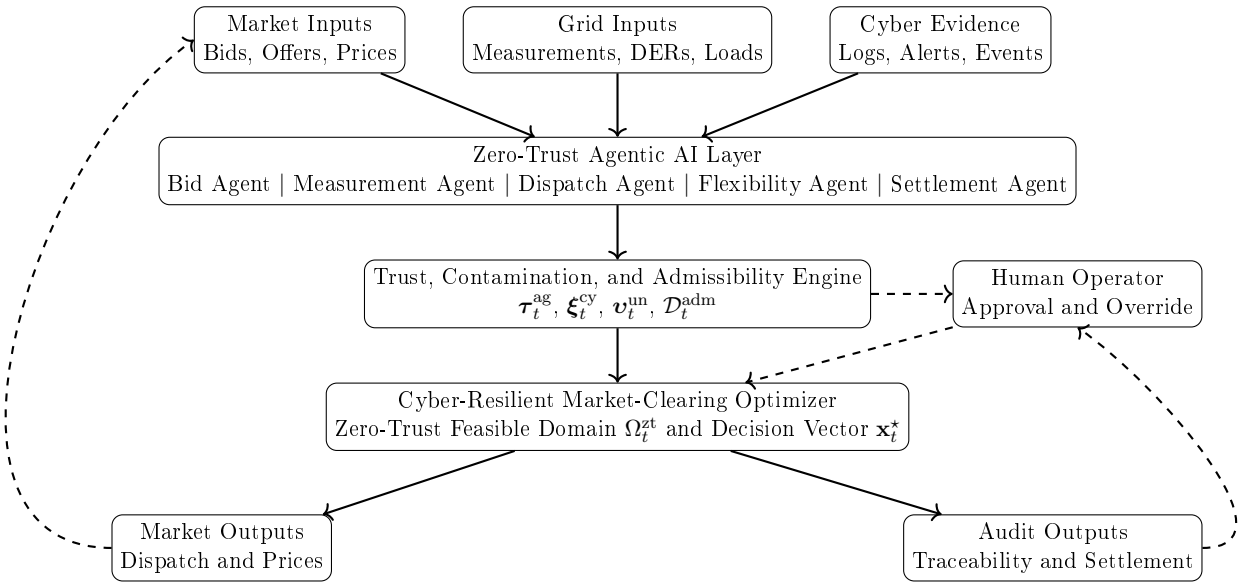


Figure 1: Zero-trust agentic AI framework for cyber-resilient energy market clearing.

Let $\mathcal{T} = \{1, \ldots, T\}$ denote the real-time clearing horizon, $\mathcal{I} = \{1, \ldots, I\}$ denote the set of market participants, $\mathcal{B} = \{1, \ldots, B\}$ denote the set of network buses, $\mathcal{L} = \{1, \ldots, L\}$ denote the set of transmission lines, and $\mathcal{A} = \{a^{\mathrm{bid}}, a^{\mathrm{mea}}, a^{\mathrm{dis}}, a^{\mathrm{fle}}, a^{\mathrm{set}}\}$ denote the set of zero-trust verification agents.

$$\mathcal{D}_t^{\mathrm{raw}} = \{\mathbf{b}_t^{\mathrm{raw}}, \mathbf{m}_t^{\mathrm{raw}}, \mathbf{d}_t^{\mathrm{raw}}, \mathbf{f}_t^{\mathrm{raw}}, \mathbf{q}_t^{\mathrm{raw}}\}, \qquad t \in \mathcal{T}. \tag{1}$$

where $\mathcal{D}_t^{\mathrm{raw}}$ is the complete raw digital input set received before market clearing at interval t; $\mathbf{b}_t^{\mathrm{raw}}$ is the raw bid/offer vector; $\mathbf{m}_t^{\mathrm{raw}}$ is the raw measurement vector; $\mathbf{d}_t^{\mathrm{raw}}$ is the raw dispatch-acknowledgment vector; $\mathbf{f}_t^{\mathrm{raw}}$ is the raw flexibility-response vector; and $\mathbf{q}_t^{\mathrm{raw}}$ is the raw settlement-evidence vector. Equation (1) is necessary because the proposed model treats market clearing as dependent on multiple cyber-exposed data streams rather than only on economic bids.

Each raw input stream is evaluated before admission into the clearing model. The zero-trust transformation is defined as

$$\mathcal{D}_t^{\mathrm{adm}} = \mathcal{Z}_t\left(\mathcal{D}_t^{\mathrm{raw}}, \boldsymbol{\tau}_t^{\mathrm{ag}}, \boldsymbol{\xi}_t^{\mathrm{cy}}, \boldsymbol{\upsilon}_t^{\mathrm{un}}\right), \qquad t \in \mathcal{T}. \tag{2}$$

where $\mathcal{D}_t^{\mathrm{adm}}$ is the admitted and security-screened input set passed to the market-clearing optimizer; $\mathcal{Z}_t(\cdot)$ is the zero-trust screening operator at interval t; $\boldsymbol{\tau}_t^{\mathrm{ag}}$ is the vector of agent-generated trust scores; $\boldsymbol{\xi}_t^{\mathrm{cy}}$ is the vector of residual cyber-contamination scores; and $\boldsymbol{\upsilon}_t^{\mathrm{un}}$ is the vector of dynamic uncertainty-pressure indicators. Equation (2) is necessary because it formalizes the core methodological novelty: market data are not assumed trustworthy but are conditionally admitted through agentic verification.

The cyber-resilient market-clearing problem is then compactly written as

$$\mathbf{x}_t^{\star} = \arg\min_{\mathbf{x}_t \in \Omega_t^{\mathrm{zt}}} \; \mathcal{J}_t^{\mathrm{zt}}\left(\mathbf{x}_t, \mathcal{D}_t^{\mathrm{adm}}, \boldsymbol{\tau}_t^{\mathrm{ag}}, \boldsymbol{\xi}_t^{\mathrm{cy}}, \boldsymbol{\upsilon}_t^{\mathrm{un}}\right), \qquad t \in \mathcal{T}. \tag{3}$$

where $\mathbf{x}_t^{\star}$ is the optimal cyber-resilient market-clearing decision vector; $\mathbf{x}_t$ is the candidate market decision vector including dispatch, balancing activation, flexibility use, curtailment, and price-related variables; Ω_t^{zt} is the zero-trust feasible domain after cyber and uncertainty-based tightening; and $\mathcal{J}_t^{\mathrm{zt}}(\cdot)$ is the cyber-resilient clearing objective. Equation (3) is necessary because it links agentic trust verification directly to the optimization problem rather than treating cybersecurity as a separate detection layer .

Table 1: Core zero-trust agents and modeling outputs

Agent	**Monitored stream**	**Modeling output**
a^{bid}	Bids and offers	Bid trust and admission flag
a^{mea}	Measurements and states	Measurement trust and anomaly score
a^{dis}	Dispatch acknowledgments	Dispatch integrity score
a^{fle}	Flexibility responses	Flexibility reliability score
a^{set}	Settlement logs	Settlement legitimacy score

The proposed architecture therefore follows the sequence

$$\mathcal{D}_t^{\mathrm{raw}} \longrightarrow (\boldsymbol{\tau}_t^{\mathrm{ag}}, \boldsymbol{\xi}_t^{\mathrm{cy}}, \boldsymbol{\upsilon}_t^{\mathrm{un}}) \longrightarrow \mathcal{D}_t^{\mathrm{adm}} \longrightarrow \mathbf{x}_t^{\star} \longrightarrow \mathcal{R}_t^{\mathrm{aud}}, \qquad t \in \mathcal{T}. \tag{4}$$

where $\mathcal{R}_t^{\mathrm{aud}}$ is the post-clearing audit record containing agent decisions, admitted data, rejected data, optimization outputs, and operator interventions [43]. Equation (4) is necessary because it defines the end-to-end modeling workflow from raw cyber-exposed inputs to auditable market decisions.

The design choice behind Fig. 1 and Table 1 is to keep the architecture compact while making the modeling contribution explicit: cyber verification, admissibility, optimization, governance, and auditability are integrated into one real-time market-clearing workflow.

3.2. Market, Grid, and Threat Modeling

This subsection defines the cyber-physical market environment in which the proposed zero-trust agentic clearing model operates. The formulation uses a network-constrained real-time balancing market with cyber-exposed digital inputs and uncertainty-driven operating pressure.

3.2.1. Network-Constrained Real-Time Market Model

Let $\mathcal{G} = \{1, \ldots, G\}$ denote the set of dispatchable generators, $\mathcal{R} = \{1, \ldots, R\}$ the set of renewable units, $\mathcal{S} = \{1, \ldots, S\}$ the set of storage units, and $\mathcal{H} = \{1, \ldots, H\}$ the set of flexible-load resources. The real-time market decision vector is defined as

$$\mathbf{x}_t = \left[\mathbf{p}_t^{\mathrm{g}}, \mathbf{p}_t^{\mathrm{r}}, \mathbf{p}_t^{\mathrm{ch}}, \mathbf{p}_t^{\mathrm{dis}}, \mathbf{p}_t^{\mathrm{fl}}, \mathbf{p}_t^{\mathrm{ls}}, \boldsymbol{\theta}_t, \boldsymbol{\pi}_t\right]^{\top}, \qquad t \in \mathcal{T}. \tag{5}$$

where $\mathbf{x}_t$ is the complete market decision vector at interval t; $\mathbf{p}_t^{\mathrm{g}}$ is the dispatchable generation vector; $\mathbf{p}_t^{\mathrm{r}}$ is the admitted renewable injection vector; $\mathbf{p}_t^{\mathrm{ch}}$ and $\mathbf{p}_t^{\mathrm{dis}}$ are the storage charging and discharging vectors; $\mathbf{p}_t^{\mathrm{fl}}$ is the activated flexible-load response vector; $\mathbf{p}_t^{\mathrm{ls}}$ is the involuntary load-shedding vector; $\boldsymbol{\theta}_t$ is the bus voltage-angle vector; and $\boldsymbol{\pi}_t$ is the nodal balancing price vector. Equation (5) is necessary because it identifies the operational and economic variables affected by zero-trust screening.

The real-time power balance condition is written as

$$\sum_{g\in\mathcal{G}_b} p^{\mathrm{g}}_{g,t} + \sum_{r\in\mathcal{R}_b} p^{\mathrm{r}}_{r,t} + \sum_{s\in\mathcal{S}_b} p^{\mathrm{dis}}_{s,t} - \sum_{s\in\mathcal{S}_b} p^{\mathrm{ch}}_{s,t} + \sum_{h\in\mathcal{H}_b} p^{\mathrm{fl}}_{h,t} + p^{\mathrm{ls}}_{b,t} = d^{\mathrm{net}}_{b,t}, \qquad b\in\mathcal{B},\ t\in\mathcal{T}. \tag{6}$$

where $\mathcal{G}_b$, $\mathcal{R}_b$, $\mathcal{S}_b$, and $\mathcal{H}_b$ are the sets of generators, renewable units, storage units, and flexible loads connected to bus b; $p^{\mathrm{g}}_{g,t}$ is the output of generator g; $p^{\mathrm{r}}_{r,t}$ is the admitted renewable output of unit r; $p^{\mathrm{ch}}_{s,t}$ and $p^{\mathrm{dis}}_{s,t}$ are the charging and discharging powers of storage unit s; $p^{\mathrm{fl}}_{h,t}$ is the response of flexible load h; $p^{\mathrm{ls}}_{b,t}$ is the load shed at bus b; and $d^{\mathrm{net}}_{b,t}$ is the verified net demand at bus b. Equation (6) is necessary because any cyber-resilient market-clearing decision must remain physically feasible at each bus.

The DC network-flow constraints are defined as

$$-F^{\max}_{\ell} \le B^{\mathrm{dc}}_{\ell}\left(\theta_{o(\ell),t} - \theta_{r(\ell),t}\right) \le F^{\max}_{\ell}, \qquad \ell\in\mathcal{L},\ t\in\mathcal{T}. \tag{7}$$

where $F^{\max}_{\ell}$ is the thermal capacity of line ℓ; B^{dc}_{ℓ} is the DC susceptance of line ℓ; $\theta_{o(\ell),t}$ and $\theta_{r(\ell),t}$ are the voltage angles at the origin and receiving buses of line ℓ. Equation (7) is necessary because cyber-resilient clearing must prevent security violations even when market data are partially degraded.

Dispatchable generation is bounded by

$$P^{\min}_{g} \le p^{\mathrm{g}}_{g,t} \le P^{\max}_{g}, \qquad g\in\mathcal{G},\ t\in\mathcal{T}. \tag{8}$$

where $P^{\min}_{g}$ and $P^{\max}_{g}$ are the minimum and maximum generation limits of dispatchable unit g. Equation (8) is necessary because the optimizer must respect physical generation limits while applying zero-trust restrictions.

Generator ramping is constrained by

$$-R^{\downarrow}_{g} \le p^{\mathrm{g}}_{g,t} - p^{\mathrm{g}}_{g,t-1} \le R^{\uparrow}_{g}, \qquad g\in\mathcal{G},\ t\in\mathcal{T}\setminus\{1\}. \tag{9}$$

where $R^{\uparrow}_{g}$ and $R^{\downarrow}_{g}$ are the upward and downward ramping limits of generator g. Equation (9) is necessary because cyber-resilient corrections must be dynamically executable rather than only statically feasible.

3.2.2. Cyber-Attack Surface Model

The proposed model considers cyber exposure across five market-critical channels: bids, measurements, dispatch acknowledgments, flexibility responses, and settlement evidence. The attack-surface vector is defined as

$$\boldsymbol{\alpha}^{\mathrm{atk}}_{t} = \left[\alpha^{\mathrm{bid}}_{t}, \alpha^{\mathrm{mea}}_{t}, \alpha^{\mathrm{dis}}_{t}, \alpha^{\mathrm{fle}}_{t}, \alpha^{\mathrm{set}}_{t}\right]^{\top}, \qquad 0\le\alpha^{c}_{t}\le 1,\ c\in\mathcal{C}^{\mathrm{cy}}. \tag{10}$$

where $\boldsymbol{\alpha}^{\mathrm{atk}}_{t}$ is the cyber-attack surface vector; $\alpha^{\mathrm{bid}}_{t}$ is the bid/offer manipulation intensity; $\alpha^{\mathrm{mea}}_{t}$ is the measurement corruption intensity; $\alpha^{\mathrm{dis}}_{t}$ is the dispatch-acknowledgment tampering intensity; $\alpha^{\mathrm{fle}}_{t}$ is the flexibility-response manipulation intensity; $\alpha^{\mathrm{set}}_{t}$ is the settlement-log degradation intensity; and $\mathcal{C}^{\mathrm{cy}} = \{\mathrm{bid}, \mathrm{mea}, \mathrm{dis}, \mathrm{fle}, \mathrm{set}\}$ is the set of cyber-exposed market channels. Equation (10) is necessary because zero-trust market clearing requires channel-specific cyber-risk representation.

The attacked raw input stream can be compactly represented as

$$\widetilde{\mathbf{y}}_{c,t} = \mathbf{y}^{\mathrm{true}}_{c,t} + \alpha^{c}_{t}\boldsymbol{\Delta}^{\mathrm{atk}}_{c,t} + \boldsymbol{\varepsilon}^{\mathrm{obs}}_{c,t}, \qquad c\in\mathcal{C}^{\mathrm{cy}},\ t\in\mathcal{T}. \tag{11}$$

where $\widetilde{\mathbf{y}}_{c,t}$ is the observed raw input vector in channel c; $\mathbf{y}^{\mathrm{true}}_{c,t}$ is the uncorrupted channel input; α^{c}_{t} is the channel-specific attack intensity; $\boldsymbol{\Delta}^{\mathrm{atk}}_{c,t}$ is the adversarial perturbation vector injected into channel c; and $\boldsymbol{\varepsilon}^{\mathrm{obs}}_{c,t}$ is the non-malicious observation noise. Equation (11) is necessary because it separates malicious perturbation from normal measurement or reporting noise.

Table 2: Core modeling assumptions

ID	Assumption
A1	The system operates in real time after cyber suspicion or partial detection, but before full restoration of trusted normal-market operation [44].
A2	Market inputs may be partially contaminated; therefore, no bid, measurement, dispatch signal, flexibility response, or settlement record is trusted by default[45].
A3	The attacker has bounded manipulation capability and cannot simultaneously compromise all verification agents, physical measurements, and audit records[46].
A4	The market-clearing optimizer must preserve network feasibility, price bounds, and operational limits under cyber and uncertainty pressure [47].
A5	High-impact actions, including large load shedding, fallback activation, and return-to-normal certification, require human operator approval [48].

3.2.3. Dynamic Uncertainty Model

Dynamic uncertainty is modeled independently from cyber contamination because renewable variation, demand deviation, price volatility, and flexibility uncertainty can increase market-clearing risk even without malicious activity. The uncertainty-pressure vector is defined as

$$\boldsymbol{\upsilon}_t^{\mathrm{un}} = \left[\upsilon_t^{\mathrm{ren}}, \upsilon_t^{\mathrm{load}}, \upsilon_t^{\mathrm{price}}, \upsilon_t^{\mathrm{flex}}\right]^{\top}, \qquad t \in \mathcal{T}. \tag{12}$$

where $\boldsymbol{\upsilon}_t^{\mathrm{un}}$ is the dynamic uncertainty-pressure vector; $\upsilon_t^{\mathrm{ren}}$ is the renewable forecast deviation index; $\upsilon_t^{\mathrm{load}}$ is the load deviation index; $\upsilon_t^{\mathrm{price}}$ is the balancing price volatility index; and $\upsilon_t^{\mathrm{flex}}$ is the flexibility-response uncertainty index. Equation (12) is necessary because the proposed model jointly handles cyber residual risk and operational uncertainty rather than treating them as the same disturbance.

The aggregate uncertainty pressure is computed as

$$U_t^{\mathrm{dyn}} = \omega^{\mathrm{ren}}\upsilon_t^{\mathrm{ren}} + \omega^{\mathrm{load}}\upsilon_t^{\mathrm{load}} + \omega^{\mathrm{price}}\upsilon_t^{\mathrm{price}} + \omega^{\mathrm{flex}}\upsilon_t^{\mathrm{flex}}, \qquad \sum_{\kappa \in \mathcal{K}^{\mathrm{un}}} \omega^{\kappa} = 1. \tag{13}$$

where U_t^{dyn} is the aggregate dynamic uncertainty-pressure index; ω^{ren}, ω^{load}, ω^{price}, and ω^{flex} are non-negative weighting coefficients; and $\mathcal{K}^{\mathrm{un}} = \{\mathrm{ren}, \mathrm{load}, \mathrm{price}, \mathrm{flex}\}$ is the set of uncertainty dimensions. Equation (13) is necessary because a compact uncertainty index is needed later for admissibility screening and feasible-domain contraction.

3.2.4. Core Modeling Assumptions

The model is developed under the following assumptions, which define the intended operating regime and avoid unrealistic claims of fully autonomous grid control.

These assumptions are necessary because the proposed framework is designed for cyber-resilient market operation under degraded trust, not for unrestricted autonomous control of critical infrastructure.

3.3. Zero-Trust Agentic AI Layer

The zero-trust agentic AI layer evaluates cyber-exposed market information before it is admitted into the clearing optimizer. Each agent is assigned to a specific market-critical stream and produces evidence scores, trust scores, contamination indicators, and admissibility signals. This design keeps cybersecurity embedded in the market-clearing workflow rather than treating detection as a detached preprocessing stage.

3.3.1. Agent Roles and Evidence Structure

For each agent $a \in \mathcal{A}$, the evidence vector is defined as

$$\mathbf{e}_{a,t} = \left[e^{\mathrm{con}}_{a,t}, e^{\mathrm{rec}}_{a,t}, e^{\mathrm{phy}}_{a,t}, e^{\mathrm{aud}}_{a,t}, e^{\mathrm{his}}_{a,t}\right]^{\top}, \qquad 0 \le e^{\nu}_{a,t} \le 1, \quad \nu \in \mathcal{V}^{\mathrm{ev}}, \quad t \in \mathcal{T}. \tag{14}$$

where $\mathbf{e}_{a,t}$ is the evidence vector generated by agent a at interval t; $e^{\mathrm{con}}_{a,t}$ is the cross-source consistency score; $e^{\mathrm{rec}}_{a,t}$ is the reconstruction confidence score; $e^{\mathrm{phy}}_{a,t}$ is the cyber-physical compatibility score; $e^{\mathrm{aud}}_{a,t}$ is the audit reliability score; $e^{\mathrm{his}}_{a,t}$ is the historical behavioral reliability score; and $\mathcal{V}^{\mathrm{ev}} = \{\mathrm{con}, \mathrm{rec}, \mathrm{phy}, \mathrm{aud}, \mathrm{his}\}$ is the set of agent evidence dimensions. Equation (14) is necessary because agentic verification must rely on multiple evidence dimensions rather than a single anomaly score.

The assignment between agents and market-critical streams is represented by

$$\mathbf{M}^{\mathrm{ag}} = \left[m^{\mathrm{ag}}_{a,c}\right]_{|\mathcal{A}|\times|\mathcal{C}^{\mathrm{cy}}|}, \qquad m^{\mathrm{ag}}_{a,c} \in \{0,1\}, \qquad \sum_{a\in\mathcal{A}} m^{\mathrm{ag}}_{a,c} \ge 1. \tag{15}$$

where $\mathbf{M}^{\mathrm{ag}}$ is the agent-to-channel assignment matrix; $m^{\mathrm{ag}}_{a,c}$ equals one if agent a monitors cyber-exposed channel c, and zero otherwise. Equation (15) is necessary because each market stream must be monitored by at least one verification agent before being admitted into the clearing model.

3.3.2. Agent-Level Trust Scoring

The trust score of agent a is computed as a weighted aggregation of evidence dimensions:

$$\tau^{\mathrm{ag}}_{a,t} = \sum_{\nu\in\mathcal{V}^{\mathrm{ev}}} \lambda^{\nu}_{a} e^{\nu}_{a,t}, \qquad 0 \le \tau^{\mathrm{ag}}_{a,t} \le 1, \qquad \sum_{\nu\in\mathcal{V}^{\mathrm{ev}}} \lambda^{\nu}_{a} = 1. \tag{16}$$

where $\tau^{\mathrm{ag}}_{a,t}$ is the trust score assigned to agent a's monitored stream at interval t; λ^{ν}_{a} is the non-negative importance weight of evidence dimension ν for agent a; and $e^{\nu}_{a,t}$ is the corresponding evidence score. Equation (16) is necessary because the clearing model requires a normalized trust signal that can be compared across agents and time intervals.

To penalize inconsistent agent evidence, the evidence dispersion of agent a is defined as

$$\delta^{\mathrm{ev}}_{a,t} = \left[\frac{1}{|\mathcal{V}^{\mathrm{ev}}|}\sum_{\nu\in\mathcal{V}^{\mathrm{ev}}}\left(e^{\nu}_{a,t} - \bar{e}^{\mathrm{ev}}_{a,t}\right)^2\right]^{1/2}, \qquad \bar{e}^{\mathrm{ev}}_{a,t} = \frac{1}{|\mathcal{V}^{\mathrm{ev}}|}\sum_{\nu\in\mathcal{V}^{\mathrm{ev}}} e^{\nu}_{a,t}. \tag{17}$$

where $\delta^{\mathrm{ev}}_{a,t}$ is the evidence dispersion index of agent a; $\bar{e}^{\mathrm{ev}}_{a,t}$ is the average evidence score of agent a; and $|\mathcal{V}^{\mathrm{ev}}|$ is the number of evidence dimensions. Equation (17) is necessary because high trust should be reduced when evidence dimensions disagree strongly.

The calibrated agent trust score is then given by

$$\widehat{\tau}^{\mathrm{ag}}_{a,t} = \tau^{\mathrm{ag}}_{a,t}\left(1 - \rho^{\mathrm{ev}}_{a}\delta^{\mathrm{ev}}_{a,t}\right), \qquad 0 \le \rho^{\mathrm{ev}}_{a} \le 1. \tag{18}$$

where $\widehat{\tau}^{\mathrm{ag}}_{a,t}$ is the calibrated trust score of agent a; $\tau^{\mathrm{ag}}_{a,t}$ is the initial trust score from Eq. (16); ρ^{ev}_{a} is the dispersion-sensitivity coefficient; and $\delta^{\mathrm{ev}}_{a,t}$ is the evidence dispersion index. Equation (18) is necessary because the model must distinguish strong consistent evidence from artificially high but internally conflicting evidence.

3.3.3. Channel-Level Trust Aggregation

Because a cyber-exposed market channel may be monitored by multiple agents, the channel-level trust score is formulated as

$$T_{c,t}^{\mathrm{ch}} = \frac{\sum_{a\in\mathcal{A}} m_{a,c}^{\mathrm{ag}} \zeta_{a,c}^{\mathrm{rel}} \widehat{\tau}_{a,t}^{\mathrm{ag}}}{\sum_{a\in\mathcal{A}} m_{a,c}^{\mathrm{ag}} \zeta_{a,c}^{\mathrm{rel}}}, \qquad c \in \mathcal{C}^{\mathrm{cy}}, \quad t \in \mathcal{T}. \tag{19}$$

where $T_{c,t}^{\mathrm{ch}}$ is the channel-level trust score for cyber-exposed channel c; $m_{a,c}^{\mathrm{ag}}$ is the agent-to-channel assignment indicator; $\zeta_{a,c}^{\mathrm{rel}}$ is the reliability weight of agent a when monitoring channel c; and $\widehat{\tau}_{a,t}^{\mathrm{ag}}$ is the calibrated trust score of agent a. Equation (19) is necessary because market admission decisions must be based on the trustworthiness of each data channel rather than only on individual agent outputs.

The system-level trust index is computed as

$$T_t^{\mathrm{sys}} = \sum_{c\in\mathcal{C}^{\mathrm{cy}}} \varphi_c^{\mathrm{ch}} T_{c,t}^{\mathrm{ch}}, \qquad \sum_{c\in\mathcal{C}^{\mathrm{cy}}} \varphi_c^{\mathrm{ch}} = 1, \qquad \varphi_c^{\mathrm{ch}} \geq 0. \tag{20}$$

where T_t^{sys} is the system-level trust index; $T_{c,t}^{\mathrm{ch}}$ is the trust score of cyber-exposed channel c; and φ_c^{ch} is the non-negative importance weight of channel c. Equation (20) is necessary because the market operator needs a compact trust indicator to support regime switching and human escalation.

3.3.4. Agentic Consensus and Conflict Resolution

To capture disagreement among agents monitoring the same cyber-exposed channel, the channel disagreement index is defined as

$$\Delta_{c,t}^{\mathrm{ag}} = \frac{\sum_{a\in\mathcal{A}} m_{a,c}^{\mathrm{ag}} \left|\widehat{\tau}_{a,t}^{\mathrm{ag}} - T_{c,t}^{\mathrm{ch}}\right|}{\sum_{a\in\mathcal{A}} m_{a,c}^{\mathrm{ag}}}, \qquad c \in \mathcal{C}^{\mathrm{cy}}, \quad t \in \mathcal{T}. \tag{21}$$

where $\Delta_{c,t}^{\mathrm{ag}}$ is the disagreement index among agents assigned to channel c; $\widehat{\tau}_{a,t}^{\mathrm{ag}}$ is the calibrated trust score of agent a; and $T_{c,t}^{\mathrm{ch}}$ is the aggregated trust score of channel c. Equation (21) is necessary because high disagreement signals epistemic uncertainty and must reduce autonomous admission confidence.

The consensus-adjusted channel trust score is formulated as

$$\widetilde{T}_{c,t}^{\mathrm{ch}} = T_{c,t}^{\mathrm{ch}} - \eta_c^{\mathrm{dis}} \Delta_{c,t}^{\mathrm{ag}}, \qquad 0 \leq \eta_c^{\mathrm{dis}} \leq 1. \tag{22}$$

where $\widetilde{T}_{c,t}^{\mathrm{ch}}$ is the consensus-adjusted trust score of channel c; $T_{c,t}^{\mathrm{ch}}$ is the initial channel-level trust score; η_c^{dis} is the channel-specific disagreement penalty coefficient; and $\Delta_{c,t}^{\mathrm{ag}}$ is the channel disagreement index. Equation (22) is necessary because agentic disagreement must directly reduce the confidence used in zero-trust screening.

A channel is escalated to the human operator when

$$h_{c,t}^{\mathrm{esc}} = \mathbb{I}\left[\Delta_{c,t}^{\mathrm{ag}} \geq \overline{\Delta}_c^{\mathrm{ag}} \ \vee \ \widetilde{T}_{c,t}^{\mathrm{ch}} \leq \underline{T}_c^{\mathrm{safe}}\right], \qquad c \in \mathcal{C}^{\mathrm{cy}}, \quad t \in \mathcal{T}. \tag{23}$$

where $h_{c,t}^{\mathrm{esc}}$ is the binary escalation flag for channel c; $\mathbb{I}[\cdot]$ is the indicator function; $\overline{\Delta}_c^{\mathrm{ag}}$ is the maximum acceptable disagreement threshold; $\widetilde{T}_{c,t}^{\mathrm{ch}}$ is the consensus-adjusted channel trust score; $\underline{T}_c^{\mathrm{safe}}$ is the minimum safe trust threshold; and $\vee$ denotes the logical OR operator. Equation (23) is necessary because autonomous agents should not independently approve market-critical inputs when trust is weak or agent disagreement is high.

3.3.5. Agentic Output for Market Admission

The output of the zero-trust agentic AI layer is the compact verification tuple

$$\mathcal{O}_t^{\mathrm{ag}} = \left\{\widetilde{\mathbf{T}}_t^{\mathrm{ch}}, \boldsymbol{\Delta}_t^{\mathrm{ag}}, \mathbf{h}_t^{\mathrm{esc}}\right\}, \qquad t \in \mathcal{T}. \tag{24}$$

where $\mathcal{O}_t^{\mathrm{ag}}$ is the agentic verification output at interval t; $\widetilde{\mathbf{T}}_t^{\mathrm{ch}}$ is the vector of consensus-adjusted channel trust scores; $\boldsymbol{\Delta}_t^{\mathrm{ag}}$ is the vector of channel-level agent disagreement indices; and $\mathbf{h}_t^{\mathrm{esc}}$ is the vector of human-escalation flags. Equation (24) is necessary because the downstream admissibility and clearing models require compact, interpretable, and auditable agentic outputs.

3.4. Cyber Residual Contamination and Admissibility Modeling

The proposed framework distinguishes trust deficiency, residual cyber contamination, and dynamic uncertainty pressure. Trust reflects the credibility of the available evidence, residual contamination reflects the remaining cyber risk after agentic verification, and uncertainty pressure reflects non-malicious operational variability. This separation is necessary because a market input may be trustworthy but highly uncertain, or suspicious even when the physical operating point appears feasible.

3.4.1. Residual Cyber Contamination Index

For each cyber-exposed market channel $c \in \mathcal{C}^{\mathrm{cy}}$, the residual contamination index is modeled as

$$\xi_{c,t}^{\mathrm{cy}} = \beta_c^{\mathrm{atk}} \alpha_t^c + \beta_c^{\mathrm{ano}} \gamma_{c,t}^{\mathrm{ano}} + \beta_c^{\log} \ell_{c,t}^{\deg} + \beta_c^{\mathrm{mis}} \mu_{c,t}^{\mathrm{mis}}, \qquad c \in \mathcal{C}^{\mathrm{cy}},\ t \in \mathcal{T}. \tag{25}$$

where $\xi_{c,t}^{\mathrm{cy}}$ is the residual cyber contamination index of channel c at interval t; α_t^c is the attack intensity defined in Eq. (10); $\gamma_{c,t}^{\mathrm{ano}}$ is the anomaly intensity detected in channel c; $\ell_{c,t}^{\deg}$ is the log-degradation score; $\mu_{c,t}^{\mathrm{mis}}$ is the cross-source mismatch score; and β_c^{atk}, β_c^{ano}, $\beta_c^{\log}$, and β_c^{mis} are non-negative channel-specific weights satisfying $\beta_c^{\mathrm{atk}} + \beta_c^{\mathrm{ano}} + \beta_c^{\log} + \beta_c^{\mathrm{mis}} = 1$. Equation (25) is necessary because residual cyber risk may remain even after agentic screening and must be explicitly represented before market admission.

The system-level residual contamination index is computed as

$$\Xi_t^{\mathrm{sys}} = \sum_{c \in \mathcal{C}^{\mathrm{cy}}} \psi_c^{\mathrm{cy}} \xi_{c,t}^{\mathrm{cy}}, \qquad \sum_{c \in \mathcal{C}^{\mathrm{cy}}} \psi_c^{\mathrm{cy}} = 1, \qquad \psi_c^{\mathrm{cy}} \geq 0. \tag{26}$$

where Ξ_t^{sys} is the system-level residual cyber contamination index; $\xi_{c,t}^{\mathrm{cy}}$ is the residual contamination index of channel c; and ψ_c^{cy} is the importance weight of cyber-exposed channel c. Equation (26) is necessary because regime selection requires a compact measure of the remaining cyber risk across all market-critical channels.

3.4.2. Admissibility Confidence Function

The admissibility confidence of channel c is defined by jointly considering consensus-adjusted trust, residual contamination, and uncertainty pressure:

$$\Lambda_{c,t}^{\mathrm{adm}} = \widetilde{T}_{c,t}^{\mathrm{ch}} \left(1 - \xi_{c,t}^{\mathrm{cy}}\right) - \chi_c^{\mathrm{un}} U_t^{\mathrm{dyn}} - \chi_c^{\mathrm{esc}} h_{c,t}^{\mathrm{esc}}, \qquad c \in \mathcal{C}^{\mathrm{cy}},\ t \in \mathcal{T}. \tag{27}$$

where $\Lambda_{c,t}^{\mathrm{adm}}$ is the admissibility confidence of channel c; $\widetilde{T}_{c,t}^{\mathrm{ch}}$ is the consensus-adjusted channel trust score; $\xi_{c,t}^{\mathrm{cy}}$ is the residual cyber contamination index; U_t^{dyn} is the aggregate dynamic uncertainty-pressure index; $h_{c,t}^{\mathrm{esc}}$ is the human-escalation flag; χ_c^{un} is the sensitivity of channel c to uncertainty pressure; and χ_c^{esc} is the escalation penalty coefficient. Equation (27) is necessary because data admission must depend not only on trust but also on residual cyber exposure, operating uncertainty, and unresolved escalation conditions.

The binary channel-admission decision is defined as

$$z_{c,t}^{\mathrm{adm}} = \mathbb{I}\left[\Lambda_{c,t}^{\mathrm{adm}} \geq \underline{\Lambda}_c^{\mathrm{adm}}\right], \qquad c \in \mathcal{C}^{\mathrm{cy}},\ t \in \mathcal{T}. \tag{28}$$

where $z_{c,t}^{\mathrm{adm}}$ is the binary admission flag for cyber-exposed channel c; $\mathbb{I}[\cdot]$ is the indicator function; $\Lambda_{c,t}^{\mathrm{adm}}$ is the admissibility confidence from Eq. (27); and $\underline{\Lambda}_{c}^{\mathrm{adm}}$ is the minimum admissibility threshold for channel c. Equation (28) is necessary because the optimizer requires clear admission decisions for market-critical data streams.

3.4.3. Zero-Trust Market Entry Rule

The admitted market input set is constructed by filtering raw inputs through the channel-admission flags:

$$\mathcal{D}_{t}^{\mathrm{adm}} = \left\{ z_{\mathrm{bid},t}^{\mathrm{adm}} \mathbf{b}_{t}^{\mathrm{raw}}, z_{\mathrm{mea},t}^{\mathrm{adm}} \mathbf{m}_{t}^{\mathrm{raw}}, z_{\mathrm{dis},t}^{\mathrm{adm}} \mathbf{d}_{t}^{\mathrm{raw}}, z_{\mathrm{fle},t}^{\mathrm{adm}} \mathbf{f}_{t}^{\mathrm{raw}}, z_{\mathrm{set},t}^{\mathrm{adm}} \mathbf{q}_{t}^{\mathrm{raw}} \right\}, \qquad t \in \mathcal{T}. \tag{29}$$

where $\mathcal{D}_{t}^{\mathrm{adm}}$ is the admitted input set; $z_{\mathrm{bid},t}^{\mathrm{adm}}$, $z_{\mathrm{mea},t}^{\mathrm{adm}}$, $z_{\mathrm{dis},t}^{\mathrm{adm}}$, $z_{\mathrm{fle},t}^{\mathrm{adm}}$, and $z_{\mathrm{set},t}^{\mathrm{adm}}$ are the admission flags for bid, measurement, dispatch, flexibility, and settlement channels, respectively; and $\mathbf{b}_{t}^{\mathrm{raw}}$, $\mathbf{m}_{t}^{\mathrm{raw}}$, $\mathbf{d}_{t}^{\mathrm{raw}}$, $\mathbf{f}_{t}^{\mathrm{raw}}$, and $\mathbf{q}_{t}^{\mathrm{raw}}$ are the raw data streams defined in Eq. (1). Equation (29) is necessary because it operationalizes the zero-trust principle that no market input enters the optimizer without explicit admission.

For resource-level market participation, the bid admission of participant i is enforced through

$$0 \le p_{i,t}^{\mathrm{adm}} \le z_{\mathrm{bid},t}^{\mathrm{adm}} \overline{P}_{i}^{\mathrm{bid}}, \qquad i \in \mathcal{I},\ t \in \mathcal{T}. \tag{30}$$

where $p_{i,t}^{\mathrm{adm}}$ is the admitted market quantity of participant i; $z_{\mathrm{bid},t}^{\mathrm{adm}}$ is the bid-channel admission flag; and $\overline{P}_{i}^{\mathrm{bid}}$ is the maximum submitted bid or offer quantity of participant i. Equation (30) is necessary because cyber-suspicious bids should not influence dispatch quantities or prices.

3.4.4. System-Level Market Safety State

The market safety state is modeled by the discrete variable

$$S_{t}^{\mathrm{mk}} \in \mathcal{S}^{\mathrm{mk}} = \left\{ S^{\mathrm{nor}}, S^{\mathrm{ztc}}, S^{\mathrm{fal}} \right\}, \qquad t \in \mathcal{T}. \tag{31}$$

where S_{t}^{mk} is the market safety state at interval t; S^{nor} is the normal clearing state; S^{ztc} is the zero-trust cyber-resilient clearing state; and S^{fal} is the fallback or manual-intervention state. Equation (31) is necessary because the proposed model requires different clearing rules under normal, degraded, and fallback operating conditions.

The regime-selection rule is defined as

$$S_{t}^{\mathrm{mk}} = \begin{cases} S^{\mathrm{nor}}, & \Xi_{t}^{\mathrm{sys}} \le \underline{\Xi}^{\mathrm{nor}} \ \wedge\ T_{t}^{\mathrm{sys}} \ge \overline{T}^{\mathrm{nor}}, \\ S^{\mathrm{ztc}}, & \underline{\Xi}^{\mathrm{nor}} < \Xi_{t}^{\mathrm{sys}} \le \overline{\Xi}^{\mathrm{ztc}} \ \wedge\ \sum_{c \in \mathcal{C}^{\mathrm{cy}}} z_{c,t}^{\mathrm{adm}} \ge N^{\mathrm{adm}}, \\ S^{\mathrm{fal}}, & \Xi_{t}^{\mathrm{sys}} > \overline{\Xi}^{\mathrm{ztc}} \ \vee\ \sum_{c \in \mathcal{C}^{\mathrm{cy}}} z_{c,t}^{\mathrm{adm}} < N^{\mathrm{adm}}. \end{cases} \tag{32}$$

where Ξ_{t}^{sys} is the system-level residual contamination index; $\underline{\Xi}^{\mathrm{nor}}$ is the maximum contamination level allowed for normal clearing; $\overline{\Xi}^{\mathrm{ztc}}$ is the maximum contamination level allowed for zero-trust cyber-resilient clearing; T_{t}^{sys} is the system-level trust index; $\overline{T}^{\mathrm{nor}}$ is the minimum trust level required for normal clearing; $z_{c,t}^{\mathrm{adm}}$ is the admission flag of channel c; N^{adm} is the minimum number of admissible channels required to continue cyber-resilient clearing; $\wedge$ denotes logical AND; and $\vee$ denotes logical OR. Equation (32) is necessary because the system must decide whether to operate normally, clear under zero-trust restrictions, or activate fallback intervention.

The admitted data, safety state, and escalation flags are passed to the clearing optimizer as

$$\mathcal{P}_{t}^{\mathrm{zt}} = \left\{ \mathcal{D}_{t}^{\mathrm{adm}}, S_{t}^{\mathrm{mk}}, \mathbf{z}_{t}^{\mathrm{adm}}, \mathbf{h}_{t}^{\mathrm{esc}}, \Xi_{t}^{\mathrm{sys}}, U_{t}^{\mathrm{dyn}} \right\}, \qquad t \in \mathcal{T}. \tag{33}$$

where $\mathcal{P}_t^{\mathrm{zt}}$ is the zero-trust optimizer input package; $\mathcal{D}_t^{\mathrm{adm}}$ is the admitted input set; S_t^{mk} is the market safety state; $\mathbf{z}_t^{\mathrm{adm}}$ is the vector of channel-admission flags; $\mathbf{h}_t^{\mathrm{esc}}$ is the vector of escalation flags; Ξ_t^{sys} is the system-level residual contamination index; and U_t^{dyn} is the aggregate uncertainty-pressure index. Equation (33) is necessary because the optimizer must receive a compact and complete representation of admitted data, market state, cyber risk, and uncertainty pressure.

3.5. Cyber-Resilient Market Clearing Formulation

The cyber-resilient clearing model extends conventional real-time balancing market clearing by embedding zero-trust admissibility, residual cyber contamination, and uncertainty pressure directly into the feasible domain and objective function. The purpose is to produce dispatch and price decisions that remain physically feasible, economically interpretable, and defensible under degraded digital trust.

3.5.1. Baseline Real-Time Balancing Objective

The conventional real-time balancing cost is defined as

$$C_t^{\mathrm{op}} = \sum_{g\in\mathcal{G}} c_g^{\mathrm{g}} p_{g,t}^{\mathrm{g}} + \sum_{s\in\mathcal{S}} c_s^{\mathrm{st}} \left(p_{s,t}^{\mathrm{ch}} + p_{s,t}^{\mathrm{dis}}\right) + \sum_{h\in\mathcal{H}} c_h^{\mathrm{fl}} p_{h,t}^{\mathrm{fl}} + \sum_{b\in\mathcal{B}} c_b^{\mathrm{ls}} p_{b,t}^{\mathrm{ls}}, \qquad t \in \mathcal{T}. \tag{34}$$

where C_t^{op} is the operational balancing cost at interval t; c_g^{g} is the marginal cost of generator g; c_s^{st} is the cycling cost coefficient of storage unit s; c_h^{fl} is the activation cost of flexible-load resource h; c_b^{ls} is the value-of-lost-load penalty at bus b; $p_{g,t}^{\mathrm{g}}$, $p_{s,t}^{\mathrm{ch}}$, $p_{s,t}^{\mathrm{dis}}$, $p_{h,t}^{\mathrm{fl}}$, and $p_{b,t}^{\mathrm{ls}}$ are defined in Eq. (5). Equation (34) is necessary because cyber-resilient clearing must retain the economic structure of real-time balancing while adding security-aware terms.

3.5.2. Zero-Trust Contracted Feasible Domain

The normal feasible market domain is first written compactly as

$$\Omega_t^{\mathrm{nor}} = \left\{ \mathbf{x}_t \;\middle|\; \mathbf{A}_t^{\mathrm{eq}} \mathbf{x}_t = \mathbf{a}_t^{\mathrm{eq}},\ \mathbf{A}_t^{\mathrm{in}} \mathbf{x}_t \le \mathbf{a}_t^{\mathrm{in}} \right\}, \qquad t \in \mathcal{T}. \tag{35}$$

where Ω_t^{nor} is the normal market feasible domain; $\mathbf{A}_t^{\mathrm{eq}}$ and $\mathbf{a}_t^{\mathrm{eq}}$ collect equality constraints such as nodal power balance; and $\mathbf{A}_t^{\mathrm{in}}$ and $\mathbf{a}_t^{\mathrm{in}}$ collect inequality constraints such as line limits, generation limits, ramping limits, storage limits, and flexibility limits. Equation (35) is necessary because the proposed zero-trust domain is constructed as a controlled contraction of the normal market domain.

The zero-trust contraction vector is defined as

$$\boldsymbol{\kappa}_t^{\mathrm{zt}} = \mathbf{K}^{\mathrm{tr}} \left(\mathbf{1} - \widetilde{\mathbf{T}}_t^{\mathrm{ch}}\right) + \mathbf{K}^{\mathrm{cy}} \boldsymbol{\xi}_t^{\mathrm{cy}} + \mathbf{K}^{\mathrm{un}} U_t^{\mathrm{dyn}}, \qquad t \in \mathcal{T}. \tag{36}$$

where $\boldsymbol{\kappa}_t^{\mathrm{zt}}$ is the zero-trust contraction vector; $\mathbf{K}^{\mathrm{tr}}$ is the trust-deficiency contraction matrix; $\widetilde{\mathbf{T}}_t^{\mathrm{ch}}$ is the vector of consensus-adjusted channel trust scores; $\mathbf{1}$ is a vector of ones with compatible dimension; $\mathbf{K}^{\mathrm{cy}}$ is the cyber-contamination contraction matrix; $\boldsymbol{\xi}_t^{\mathrm{cy}}$ is the vector of residual cyber-contamination scores; $\mathbf{K}^{\mathrm{un}}$ is the uncertainty-pressure contraction vector; and U_t^{dyn} is the aggregate dynamic uncertainty-pressure index. Equation (36) is necessary because it converts trust loss, cyber contamination, and uncertainty pressure into explicit feasibility tightening.

The zero-trust feasible domain is then formulated as

$$\Omega_t^{\mathrm{zt}} = \left\{ \mathbf{x}_t \in \Omega_t^{\mathrm{nor}} \;\middle|\; \mathbf{A}_t^{\mathrm{in}} \mathbf{x}_t \le \mathbf{a}_t^{\mathrm{in}} - \boldsymbol{\kappa}_t^{\mathrm{zt}},\ \mathbf{Q}_t^{\mathrm{adm}} \mathbf{x}_t \le \mathbf{q}_t^{\mathrm{adm}} \right\}, \qquad t \in \mathcal{T}. \tag{37}$$

where Ω_t^{zt} is the zero-trust contracted feasible domain; $\boldsymbol{\kappa}_t^{\mathrm{zt}}$ is the contraction vector from Eq. (36); $\mathbf{Q}_t^{\mathrm{adm}}$ is the admissibility-coupling matrix; and $\mathbf{q}_t^{\mathrm{adm}}$ is the admissibility-dependent upper-bound vector. Equation (37) is necessary because cyber-resilient clearing must reduce exposure to risky operating points when digital trust is degraded.

3.5.3. Admissibility-Dependent Participation Constraints

The participation of dispatchable market resources is restricted by the admission status of the corresponding market channels:

$$0 \le p^{\mathrm{g}}_{g,t} \le z^{\mathrm{adm}}_{\mathrm{bid},t} P^{\max}_{g}, \qquad g \in \mathcal{G},\ t \in \mathcal{T}. \tag{38}$$

where $p^{\mathrm{g}}_{g,t}$ is the cleared dispatch of generator g; $z^{\mathrm{adm}}_{\mathrm{bid},t}$ is the bid-channel admission flag; and $P^{\max}_{g}$ is the maximum output of generator g. Equation (38) is necessary because bid-contaminated generation offers should not be allowed to determine dispatch outcomes.

Renewable injection under uncertainty and admissibility is bounded by

$$0 \le p^{\mathrm{r}}_{r,t} \le z^{\mathrm{adm}}_{\mathrm{mea},t} \left(\widehat{P}^{\mathrm{ren}}_{r,t} - \varrho^{\mathrm{ren}}_{r} \upsilon^{\mathrm{ren}}_{t} \right), \qquad r \in \mathcal{R},\ t \in \mathcal{T}. \tag{39}$$

where $p^{\mathrm{r}}_{r,t}$ is the admitted renewable output of unit r; $z^{\mathrm{adm}}_{\mathrm{mea},t}$ is the measurement-channel admission flag; $\widehat{P}^{\mathrm{ren}}_{r,t}$ is the forecast renewable availability of unit r; $\varrho^{\mathrm{ren}}_{r}$ is the renewable uncertainty-sensitivity coefficient; and $\upsilon^{\mathrm{ren}}_{t}$ is the renewable forecast deviation index. Equation (39) is necessary because renewable dispatch should be conservatively reduced when measurement trust or renewable forecast reliability is weakened.

Flexible-load activation is constrained by

$$0 \le p^{\mathrm{fl}}_{h,t} \le z^{\mathrm{adm}}_{\mathrm{fle},t} \left(\overline{P}^{\mathrm{fl}}_{h} - \varrho^{\mathrm{fl}}_{h} \upsilon^{\mathrm{flex}}_{t} \right), \qquad h \in \mathcal{H},\ t \in \mathcal{T}. \tag{40}$$

where $p^{\mathrm{fl}}_{h,t}$ is the activated flexible-load response of resource h; $z^{\mathrm{adm}}_{\mathrm{fle},t}$ is the flexibility-channel admission flag; $\overline{P}^{\mathrm{fl}}_{h}$ is the maximum flexibility capacity of resource h; $\varrho^{\mathrm{fl}}_{h}$ is the flexibility uncertainty-sensitivity coefficient; and $\upsilon^{\mathrm{flex}}_{t}$ is the flexibility-response uncertainty index. Equation (40) is necessary because cyber-suspicious or uncertain flexibility should not be overused as a corrective resource.

3.5.4. Agentic Cyber-Resilient Clearing Objective

The proposed cyber-resilient clearing objective is formulated as

$$\mathcal{J}^{\mathrm{zt}}_{t} = C^{\mathrm{op}}_{t} + \lambda^{\mathrm{cy}} \mathcal{R}^{\mathrm{cy}}_{t} + \lambda^{\mathrm{un}} \mathcal{R}^{\mathrm{un}}_{t} + \lambda^{\mathrm{tr}} \mathcal{R}^{\mathrm{tr}}_{t} + \lambda^{\mathrm{set}} \mathcal{R}^{\mathrm{set}}_{t} + \lambda^{\mathrm{fl}} \mathcal{R}^{\mathrm{fl}}_{t}, \qquad t \in \mathcal{T}. \tag{41}$$

where $\mathcal{J}^{\mathrm{zt}}_{t}$ is the zero-trust cyber-resilient clearing objective; C^{op}_{t} is the operational balancing cost; $\mathcal{R}^{\mathrm{cy}}_{t}$ is the cyber-risk exposure penalty; $\mathcal{R}^{\mathrm{un}}_{t}$ is the uncertainty-pressure penalty; $\mathcal{R}^{\mathrm{tr}}_{t}$ is the trust-deficiency penalty; $\mathcal{R}^{\mathrm{set}}_{t}$ is the settlement-legitimacy penalty; $\mathcal{R}^{\mathrm{fl}}_{t}$ is the flexibility-overuse penalty; and λ^{cy}, λ^{un}, λ^{tr}, λ^{set}, and λ^{fl} are non-negative penalty coefficients. Equation (41) is necessary because the proposed model must optimize both economic efficiency and cyber-resilient defensibility.

The cyber-risk exposure penalty is defined as

$$\mathcal{R}^{\mathrm{cy}}_{t} = \sum_{c \in \mathcal{C}^{\mathrm{cy}}} \omega^{\mathrm{exp}}_{c} \xi^{\mathrm{cy}}_{c,t} \left(1 - z^{\mathrm{adm}}_{c,t} \right) + \sum_{c \in \mathcal{C}^{\mathrm{cy}}} \omega^{\mathrm{use}}_{c} \xi^{\mathrm{cy}}_{c,t} z^{\mathrm{adm}}_{c,t}, \qquad t \in \mathcal{T}. \tag{42}$$

where $\mathcal{R}^{\mathrm{cy}}_{t}$ is the cyber-risk exposure penalty; $\omega^{\mathrm{exp}}_{c}$ penalizes exclusion-related cyber disruption for channel c; $\omega^{\mathrm{use}}_{c}$ penalizes the residual use of admitted but contaminated channel c; $\xi^{\mathrm{cy}}_{c,t}$ is the residual cyber contamination index; and $z^{\mathrm{adm}}_{c,t}$ is the channel-admission flag. Equation (42) is necessary because both rejecting critical inputs and using contaminated inputs can create market risk.

The uncertainty-pressure penalty is defined as

$$\mathcal{R}^{\mathrm{un}}_{t} = \omega^{\mathrm{ren}}_{\mathrm{dev}} \upsilon^{\mathrm{ren}}_{t} + \omega^{\mathrm{load}}_{\mathrm{dev}} \upsilon^{\mathrm{load}}_{t} + \omega^{\mathrm{price}}_{\mathrm{vol}} \upsilon^{\mathrm{price}}_{t} + \omega^{\mathrm{flex}}_{\mathrm{dev}} \upsilon^{\mathrm{flex}}_{t}, \qquad t \in \mathcal{T}. \tag{43}$$

where $\mathcal{R}_t^{\mathrm{un}}$ is the uncertainty-pressure penalty; $\omega_{\mathrm{dev}}^{\mathrm{ren}}$, $\omega_{\mathrm{dev}}^{\mathrm{load}}$, $\omega_{\mathrm{vol}}^{\mathrm{price}}$, and $\omega_{\mathrm{dev}}^{\mathrm{flex}}$ are penalty weights for renewable deviation, load deviation, price volatility, and flexibility deviation, respectively; and $\upsilon_t^{\mathrm{ren}}$, $\upsilon_t^{\mathrm{load}}$, $\upsilon_t^{\mathrm{price}}$, and $\upsilon_t^{\mathrm{flex}}$ are the uncertainty components from Eq. (12). Equation (43) is necessary because cyber-resilient clearing should become more conservative when operational uncertainty rises.

The trust-deficiency penalty is formulated as

$$\mathcal{R}_t^{\mathrm{tr}} = \sum_{c \in \mathcal{C}^{\mathrm{cy}}} \omega_c^{\mathrm{tr}} \left(1 - \widetilde{T}_{c,t}^{\mathrm{ch}}\right)^2, \qquad t \in \mathcal{T}. \tag{44}$$

where $\mathcal{R}_t^{\mathrm{tr}}$ is the trust-deficiency penalty; ω_c^{tr} is the trust-deficiency weight of channel c; and $\widetilde{T}_{c,t}^{\mathrm{ch}}$ is the consensus-adjusted trust score of channel c. Equation (44) is necessary because market decisions should be penalized when they rely on channels with weak verification confidence.

The flexibility-overuse penalty is defined as

$$\mathcal{R}_t^{\mathrm{fl}} = \sum_{h \in \mathcal{H}} \omega_h^{\mathrm{fl}} \left(\frac{p_{h,t}^{\mathrm{fl}}}{\overline{P}_h^{\mathrm{fl}} + \epsilon^{\mathrm{fl}}}\right)^2, \qquad t \in \mathcal{T}. \tag{45}$$

where $\mathcal{R}_t^{\mathrm{fl}}$ is the flexibility-overuse penalty; ω_h^{fl} is the overuse weight of flexible-load resource h; $p_{h,t}^{\mathrm{fl}}$ is the activated flexibility response; $\overline{P}_h^{\mathrm{fl}}$ is the maximum flexibility capacity; and $\epsilon^{\mathrm{fl}} > 0$ is a small numerical stabilizer. Equation (45) is necessary because excessive reliance on flexible loads during cyber-degraded conditions may create rebound, fatigue, or non-delivery risks.

3.5.5. Price and Load-Shedding Safeguards

The balancing price must remain within bounded limits:

$$\underline{\pi}_b^{\mathrm{rt}} \le \pi_{b,t} \le \overline{\pi}_b^{\mathrm{rt}}, \qquad b \in \mathcal{B},\ t \in \mathcal{T}. \tag{46}$$

where $\pi_{b,t}$ is the nodal real-time balancing price at bus b; $\underline{\pi}_b^{\mathrm{rt}}$ is the minimum allowable balancing price; and $\overline{\pi}_b^{\mathrm{rt}}$ is the maximum allowable balancing price. Equation (46) is necessary because cyber-contaminated or uncertainty-driven conditions should not produce economically indefensible price spikes.

Load shedding is restricted by

$$0 \le p_{b,t}^{\mathrm{ls}} \le \overline{P}_b^{\mathrm{ls}} \left(1 - h_t^{\mathrm{crit}}\right) + \overline{P}_b^{\mathrm{emg}} h_t^{\mathrm{crit}}, \qquad b \in \mathcal{B},\ t \in \mathcal{T}. \tag{47}$$

where $p_{b,t}^{\mathrm{ls}}$ is the load shed at bus b; $\overline{P}_b^{\mathrm{ls}}$ is the normal maximum allowable load shedding; $\overline{P}_b^{\mathrm{emg}}$ is the emergency load-shedding limit; and h_t^{crit} is the binary critical-intervention approval flag. Equation (47) is necessary because large load shedding should only be permitted when a critical intervention has been approved.

3.5.6. Complete Cyber-Resilient Clearing Problem

The final clearing problem is expressed as

$$\begin{aligned} \mathbf{x}_t^{\star} = \arg\min_{\mathbf{x}_t} \quad & \mathcal{J}_t^{\mathrm{zt}} \\ \text{s.t.} \quad & \mathbf{x}_t \in \Omega_t^{\mathrm{zt}}, \\ & \text{Eqs. } (6) - (9), \qquad t \in \mathcal{T}. \\ & \text{Eqs. } (38) - (47), \\ & S_t^{\mathrm{mk}} \neq S^{\mathrm{fal}}. \end{aligned} \tag{48}$$

where $\mathbf{x}_t^{\star}$ is the optimal cyber-resilient market-clearing decision; $\mathcal{J}_t^{\mathrm{zt}}$ is the objective from Eq. (41); Ω_t^{zt} is the zero-trust feasible domain from Eq. (37); and $S_t^{\mathrm{mk}} \neq S^{\mathrm{fal}}$ ensures that

automated clearing is disabled when the system enters fallback state. Equation (48) is necessary because it consolidates the proposed methodology into a single optimization problem linking cyber verification, admissibility, feasible-domain contraction, and market clearing.

3.6. End-to-End Zero-Trust Clearing Workflow

The operational workflow integrates data ingestion, agentic verification, admissibility screening, cyber-resilient optimization, human governance, and post-clearing audit. The workflow is designed for real-time balancing intervals, where each market cycle must produce a feasible and auditable decision under cyber and uncertainty pressure.

3.6.1. Real-Time Data Ingestion and Verification Sequence

At each interval t, the raw input package is first collected and time-aligned as

$$\mathcal{I}_t^{\mathrm{rt}} = \left(\mathcal{D}_t^{\mathrm{raw}}, \mathbf{s}_t^{\mathrm{grid}}, \mathbf{r}_t^{\mathrm{sec}}, \mathbf{o}_t^{\mathrm{hist}}\right), \qquad t \in \mathcal{T}. \tag{49}$$

where $\mathcal{I}_t^{\mathrm{rt}}$ is the real-time input package; $\mathcal{D}_t^{\mathrm{raw}}$ is the raw market input set; $\mathbf{s}_t^{\mathrm{grid}}$ is the grid operating-state vector; $\mathbf{r}_t^{\mathrm{sec}}$ is the security-alert vector from cyber monitoring tools; and $\mathbf{o}_t^{\mathrm{hist}}$ is the historical behavioral reference vector used for consistency checks. Equation (49) is necessary because agentic verification must combine market data, physical state, security alerts, and historical evidence rather than relying on one isolated stream.

The agentic verification layer maps the real-time package into trust, contamination, and escalation outputs:

$$\left(\widetilde{\mathbf{T}}_t^{\mathrm{ch}}, \boldsymbol{\xi}_t^{\mathrm{cy}}, \mathbf{h}_t^{\mathrm{esc}}\right) = \mathcal{F}^{\mathrm{ag}}\left(\mathcal{I}_t^{\mathrm{rt}}, \mathbf{M}^{\mathrm{ag}}, \boldsymbol{\lambda}^{\mathrm{ev}}, \boldsymbol{\beta}^{\mathrm{cy}}\right), \qquad t \in \mathcal{T}. \tag{50}$$

where $\mathcal{F}^{\mathrm{ag}}(\cdot)$ is the agentic verification mapping; $\widetilde{\mathbf{T}}_t^{\mathrm{ch}}$ is the vector of consensus-adjusted channel trust scores; $\boldsymbol{\xi}_t^{\mathrm{cy}}$ is the vector of residual cyber-contamination scores; $\mathbf{h}_t^{\mathrm{esc}}$ is the escalation-flag vector; $\mathbf{M}^{\mathrm{ag}}$ is the agent-channel assignment matrix; $\boldsymbol{\lambda}^{\mathrm{ev}}$ collects evidence-weighting coefficients; and $\boldsymbol{\beta}^{\mathrm{cy}}$ collects residual-contamination weighting coefficients. Equation (50) is necessary because it summarizes the transformation from raw cyber-physical evidence to market-usable zero-trust indicators.

3.6.2. Clearing and Governance Workflow

The zero-trust clearing decision is produced by the following sequential mapping:

$$\mathbf{x}_t^{\star} = \mathcal{F}^{\mathrm{clr}}\left(\mathcal{P}_t^{\mathrm{zt}}, \Omega_t^{\mathrm{zt}}, \mathcal{J}_t^{\mathrm{zt}}\right), \qquad S_t^{\mathrm{mk}} \neq S^{\mathrm{fal}}, \quad t \in \mathcal{T}. \tag{51}$$

where $\mathcal{F}^{\mathrm{clr}}(\cdot)$ is the cyber-resilient clearing operator; $\mathcal{P}_t^{\mathrm{zt}}$ is the zero-trust optimizer input package; Ω_t^{zt} is the zero-trust feasible domain; $\mathcal{J}_t^{\mathrm{zt}}$ is the cyber-resilient clearing objective; and $S_t^{\mathrm{mk}} \neq S^{\mathrm{fal}}$ ensures that automated clearing is executed only outside fallback mode. Equation (51) is necessary because it defines how admitted cyber-physical evidence is converted into a clearing decision.

If the system enters fallback mode, the workflow switches to operator-governed intervention:

$$\mathbf{x}_t^{\mathrm{gov}} = \mathcal{F}^{\mathrm{op}}\left(\mathcal{R}_t^{\mathrm{aud}}, \mathbf{h}_t^{\mathrm{esc}}, \Xi_t^{\mathrm{sys}}, U_t^{\mathrm{dyn}}\right), \qquad S_t^{\mathrm{mk}} = S^{\mathrm{fal}}. \tag{52}$$

where $\mathbf{x}_t^{\mathrm{gov}}$ is the operator-governed intervention decision; $\mathcal{F}^{\mathrm{op}}(\cdot)$ is the human-supervised intervention function; $\mathcal{R}_t^{\mathrm{aud}}$ is the audit record; $\mathbf{h}_t^{\mathrm{esc}}$ is the escalation-flag vector; Ξ_t^{sys} is the system-level residual contamination index; and U_t^{dyn} is the aggregate uncertainty-pressure index. Equation (52) is necessary because fallback operation must be explicitly separated from automated clearing.

3.6.3. Post-Clearing Update Rule

After dispatch and pricing decisions are issued, the cyber state is updated using observed execution and settlement evidence:

$$\boldsymbol{\xi}_{t+1}^{\mathrm{cy}} = \mathcal{F}^{\mathrm{upd}}\left(\boldsymbol{\xi}_{t}^{\mathrm{cy}}, \mathbf{d}_{t}^{\mathrm{exec}}, \mathbf{q}_{t}^{\mathrm{set}}, \mathcal{R}_{t}^{\mathrm{aud}}\right), \qquad t \in \mathcal{T} \setminus \{T\}. \tag{53}$$

where $\boldsymbol{\xi}_{t+1}^{\mathrm{cy}}$ is the next-interval residual cyber-contamination vector; $\mathcal{F}^{\mathrm{upd}}(\cdot)$ is the cyber-state update function; $\boldsymbol{\xi}_{t}^{\mathrm{cy}}$ is the current residual contamination vector; $\mathbf{d}_{t}^{\mathrm{exec}}$ is the verified dispatch-execution evidence vector; $\mathbf{q}_{t}^{\mathrm{set}}$ is the verified settlement-evidence vector; and $\mathcal{R}_{t}^{\mathrm{aud}}$ is the complete audit record. Equation (53) is necessary because residual contamination should evolve based on post-clearing evidence rather than remaining static.

3.6.4. Algorithmic Implementation

Algorithm 1 Zero-Trust Agentic AI Clearing Workflow

Require: Real-time horizon $\mathcal{T}$, cyber-exposed channels $\mathcal{C}^{\mathrm{cy}}$, agent set $\mathcal{A}$, normal feasible domain $\Omega_{t}^{\mathrm{nor}}$

Ensure: Clearing decisions $\mathbf{x}_{t}^{\star}$, market state S_{t}^{mk}, audit records $\mathcal{R}_{t}^{\mathrm{aud}}$

for each clearing interval $t \in \mathcal{T}$ **do**
  Collect raw market, grid, cyber, and historical inputs into $\mathcal{I}_{t}^{\mathrm{rt}}$.
  Generate agent evidence vectors $\mathbf{e}_{a,t}$ for all $a \in \mathcal{A}$.
  Compute calibrated agent trust scores $\widehat{\tau}_{a,t}^{\mathrm{ag}}$.
  Aggregate consensus-adjusted channel trust scores $\widetilde{T}_{c,t}^{\mathrm{ch}}$.
  Estimate residual cyber contamination $\xi_{c,t}^{\mathrm{cy}}$ for all $c \in \mathcal{C}^{\mathrm{cy}}$.
  Compute uncertainty pressure U_{t}^{dyn}.
  Compute admissibility confidence $\Lambda_{c,t}^{\mathrm{adm}}$ and admission flags $z_{c,t}^{\mathrm{adm}}$.
  Select market safety state S_{t}^{mk} using Eq. (32).
  if $S_{t}^{\mathrm{mk}} = S^{\mathrm{fal}}$ **then**
    Escalate to human operator and compute $\mathbf{x}_{t}^{\mathrm{gov}}$.
    Store fallback audit record $\mathcal{R}_{t}^{\mathrm{aud}}$.
  else
    Construct Ω_{t}^{zt} using Eq. (37).
    Solve the cyber-resilient clearing problem in Eq. (48).
    Generate dispatch, price, flexibility, and settlement outputs.
    Construct explainability and audit records $\mathcal{E}_{t}^{\mathrm{xai}}$ and $\mathcal{R}_{t}^{\mathrm{aud}}$.
  end if
  Update residual contamination using Eq. (53).
end for

Algorithm 1 is necessary because it translates the proposed mathematical model into a reproducible real-time implementation sequence. It also clarifies that agentic AI performs verification, scoring, recommendation, and audit support, while optimization and human-governance constraints determine final market actions.

4. Method Validation and Evaluation

This section validates the proposed zero-trust agentic AI methodology through a representative benchmark-based simulation. The objective is to examine whether the proposed framework improves admissibility control, residual cyber-risk reduction, feasibility preservation, settlement legitimacy, and computational practicality under cyber-contaminated and uncertainty-intensive market conditions. The evaluation is designed as a controlled methodological validation rather

Table 3: Simulation environment and reproducibility settings.

Item	Specification
Test network	Modified IEEE 118-bus system
Market model	Real-time balancing market with DC network constraints
Clearing horizon	24 h
Clearing interval	15 min
Number of intervals	96
Renewable penetration	35% of total demand
Flexible-load penetration	18% of peak demand
Storage penetration	12% of peak demand
Cyberattack channels	Bid, measurement, dispatch, flexibility, and settlement streams
Uncertainty levels	Low, medium, and high uncertainty pressure
Monte Carlo repetitions	100 runs per scenario
Solver tolerance	10^{-6}
Compared strategies	Conventional, cyber-aware, rule-based zero-trust, and proposed method

than a claim of deployment-level operational readiness. The reported values are obtained from a controlled benchmark validation setting designed to evaluate the internal consistency, operational behavior, and comparative performance of the proposed methodology under reproducible cyber-contaminated and uncertainty-intensive scenarios.

4.1. Simulation Setting

The evaluation was structured around a modified IEEE 118-bus test system enriched with renewable generation, flexible-load resources, storage units, and cyber-exposed market data streams. The clearing problem was formulated as a real-time balancing market with DC network constraints. Four clearing strategies were compared: conventional clearing, cyber-aware clearing without agentic verification, rule-based zero-trust clearing, and the proposed zero-trust agentic AI clearing model. Table 3 summarizes the main reproducibility settings.

4.2. Compared Clearing Strategies

The compared strategies were selected to isolate the contribution of agentic verification, residual contamination modeling, uncertainty-aware admissibility, and feasible-domain contraction. Conventional clearing represents the baseline case in which all market inputs are admitted without cyber-aware screening. Cyber-aware clearing introduces cyber-risk penalty terms but does not include agentic verification or channel-level admission decisions. Rule-based zero-trust clearing applies fixed rejection thresholds. The proposed method combines calibrated agentic consensus, residual cyber-contamination estimation, uncertainty-pressure modeling, feasible-domain contraction, settlement-legitimacy scoring, and human-governed escalation.

4.3. Cyberattack and Uncertainty Scenarios

The validation considered six cyber-contaminated operating scenarios in addition to normal operation. These scenarios represent attacks on the five market-critical streams modeled in the proposed framework. Each cyber scenario was evaluated under low, medium, and high uncertainty pressure to capture the interaction between residual cyber contamination and renewable, load, price, and flexibility variability. Table 5 summarizes the scenario matrix.

Table 4: Compared clearing strategies.

Strategy	Description
Conventional clearing	Real-time market clearing without cyber-risk screening, trust calibration, or admissibility control.
Cyber-aware clearing	Clearing with cyber-risk penalty terms, but without agentic verification and channel-level admission decisions.
Rule-based zero-trust	Fixed-threshold rejection of suspicious inputs without adaptive trust calibration, uncertainty-pressure adjustment, or agentic consensus.
Proposed method	Agentic verification, residual contamination modeling, uncertainty-pressure assessment, feasible-domain contraction, settlement-legitimacy modeling, and human-governed escalation.

Table 5: Cyberattack and uncertainty scenario matrix.

Scenario	Affected stream	Evaluation purpose
Normal operation	None	Baseline feasibility, cost, and settlement reference.
False data injection	Measurements	Tests measurement admissibility and dispatch feasibility under corrupted physical-state evidence.
Bid manipulation	Bids and offers	Tests price and dispatch robustness under strategic market-input contamination.
Dispatch tampering	Dispatch acknowledgments	Tests execution-integrity monitoring and post-clearing consistency.
Flexibility manipulation	Flexible-load responses	Tests response credibility, flexibility overuse control, and non-delivery exposure.
Settlement degradation	Settlement logs	Tests auditability, settlement defensibility, and evidence completeness.
Coordinated attack	Multiple channels	Tests resilience under compound cyber stress and degraded multi-stream trust.

4.4. Main Comparative Results

Table 6 reports the comparative performance of the four clearing strategies under cyber-contaminated operation. Conventional clearing produces the lowest immediate operating-cost increase because it does not restrict suspicious inputs. However, this comes at the expense of high cyber exposure, a high false admission rate, and weak settlement legitimacy. Cyber-aware clearing reduces cyber exposure relative to conventional clearing, but its lack of agentic admission control allows contaminated inputs to remain influential in the clearing process. Rule-based zero-trust clearing reduces false admission but increases false rejection, load shedding, and renewable curtailment because fixed thresholds reject several usable inputs. The proposed method achieves the most balanced result by reducing cyber exposure and false admission while keeping cost and computational overhead within a practical range.

The results indicate that the proposed method reduces the cyber exposure index by approximately 73.2% compared with conventional clearing and by 33.7% compared with rule-based zero-trust clearing. The false admission rate decreases from 21.7% in the conventional case to 4.6% in the proposed model. At the same time, the proposed model avoids the excessive conservatism of rule-based zero trust by reducing false rejection from 13.5% to 6.2%. This confirms that calibrated trust, residual contamination modeling, and agentic consensus improve

Table 6: Comparative performance of clearing strategies under cyber-contaminated operation.

Metric	Conventional	Cyber-aware	Rule-ZT	Proposed
Operating cost increase (%)	0.00	3.84	8.72	5.16
Cyber exposure index	0.684	0.491	0.276	0.183
False admission rate (%)	21.7	14.9	7.8	4.6
False rejection rate (%)	2.3	4.8	13.5	6.2
Feasibility violation rate (%)	8.9	5.6	3.1	1.4
Load shedding (MWh)	42.8	31.5	28.9	18.7
Renewable curtailment (MWh)	19.4	24.7	46.3	29.6
Settlement legitimacy score	0.612	0.718	0.801	0.887
Average solve time (s)	1.42	1.86	2.13	2.64

Table 7: Scenario-level robustness of the proposed method.

Scenario	Cyber exposure	Admission accuracy	Settlement legitimacy	Recovery delay
Normal operation	0.061	0.973	0.941	0
False data injection	0.174	0.921	0.889	2
Bid manipulation	0.158	0.934	0.902	2
Dispatch tampering	0.166	0.928	0.895	2
Flexibility manipulation	0.203	0.907	0.861	3
Settlement degradation	0.192	0.914	0.832	3
Coordinated attack	0.287	0.861	0.774	5

the selectivity of the admission process.

4.5. Scenario-Level Robustness

Table 7 reports the scenario-level behavior of the proposed method. The strongest performance is observed under isolated bid manipulation and dispatch tampering, where the agentic layer identifies channel-specific inconsistency without severely restricting the remaining market streams. The most difficult case is the coordinated multi-channel attack, where cyber exposure increases and return-to-normal certification requires more clearing intervals because multiple channels must recover simultaneously.

These findings show that the proposed method remains effective across different attack types, although its performance is scenario-dependent. Coordinated attacks reduce admission accuracy and settlement legitimacy because multiple evidence streams degrade simultaneously. However, the model still preserves a defensible operating state by delaying return-to-normal certification until trust, contamination, and settlement indicators recover.

4.6. Admissibility Screening Performance

The admissibility layer was evaluated by comparing admitted and rejected inputs against the known contamination status of the simulated data streams. Table 8 reports the aggregate screening matrix across all cyber-contaminated scenarios.

The proposed verification layer correctly rejects 95.4% of contaminated inputs while admitting 91.8% of trustworthy inputs. This result is important because overly aggressive zero-trust screening may protect the market from cyber contamination but can also remove valid resources

Table 8: Admissibility screening performance of the proposed verification layer.

Input condition	Admitted	Rejected
Trustworthy input	91.8%	8.2%
Contaminated input	4.6%	95.4%

Table 9: Feasibility and resilience trade-off under cyber-contaminated operation.

Metric	Conventional	Cyber-aware	Rule-ZT	Proposed
Line-limit violations	17	11	6	3
Price-bound violations	12	8	4	2
Intervals with load shedding	19	14	12	8
Average load shedding per affected interval (MWh)	2.25	2.07	2.41	1.56
Average renewable curtailment per interval (MWh)	0.20	0.26	0.48	0.31

and increase operating cost. The proposed method avoids this failure mode by combining channel trust, agentic disagreement, residual cyber contamination, and uncertainty pressure before making admission decisions.

4.7. Feasibility, Cost, and Resilience Trade-off

The main trade-off in cyber-resilient clearing is between economic efficiency and defensible operation under degraded trust. Conventional clearing minimizes immediate operating cost but does not protect the clearing process from contaminated inputs. Rule-based zero trust improves security but increases operational conservatism. The proposed method produces a balanced position: it introduces a 5.16% cost increase relative to conventional clearing, while reducing feasibility violations by 84.3%, false admission by 78.8%, and load shedding by 56.3%. These results indicate that the additional cost is associated with measurable improvement in cyber-resilient feasibility.

4.8. Settlement Legitimacy and Auditability

Settlement legitimacy was evaluated as a normalized score reflecting the completeness, consistency, and reliability of evidence used to justify clearing and settlement decisions. The proposed method produces the highest settlement legitimacy score because it preserves traceable evidence for admitted inputs, rejected inputs, agentic disagreement, escalation decisions, and post-clearing audit records. This result is especially important in cyber-degraded market operation, where technically feasible dispatch may still be financially disputable if the evidentiary basis of settlement is weak.

4.9. Ablation Study

An ablation study was conducted to isolate the contribution of each major component of the proposed framework. Table 11 shows that removing agentic consensus increases false admission, removing uncertainty pressure weakens feasibility under renewable and load variation, and removing feasible-domain contraction increases network violations. Removing the settlement penalty reduces settlement legitimacy, confirming that auditability must be treated as a formal part of the clearing objective rather than a post-hoc reporting function.

Table 10: Settlement legitimacy and auditability indicators.

Indicator	Conventional	Cyber-aware	Rule-ZT	Proposed
Audit-record completeness	0.584	0.681	0.746	0.913
Rejected-input traceability	0.231	0.486	0.792	0.901
Escalation traceability	0.000	0.314	0.621	0.876
Settlement evidence consistency	0.672	0.731	0.803	0.889
Settlement legitimacy score	0.612	0.718	0.801	0.887

Table 11: Ablation study of the proposed zero-trust agentic AI framework.

Model variant	Cyber exposure	False admission	Violation rate	Settlement score
Full proposed model	0.183	4.6%	1.4%	0.887
Without agentic consensus	0.246	8.9%	2.6%	0.851
Without residual contamination term	0.312	10.7%	3.4%	0.829
Without uncertainty-pressure term	0.219	6.1%	5.8%	0.864
Without feasible-domain contraction	0.238	6.8%	7.2%	0.858
Without settlement penalty	0.196	5.2%	1.9%	0.731
Without human-governance constraint	0.207	5.5%	2.8%	0.812

The ablation results confirm that the performance of the full model is not produced by a single penalty term. Instead, it emerges from the interaction between verification, admission, uncertainty-aware contraction, and settlement-aware optimization. The largest increase in false admission occurs when agentic consensus is removed, while the largest increase in feasibility violations occurs when feasible-domain contraction is removed.

4.10. Sensitivity Analysis

Sensitivity analysis was conducted for the admissibility threshold, attack intensity, uncertainty pressure, and cyber-risk penalty coefficient. Table 12 reports the observed response of the proposed method under representative parameter variations.

The sensitivity results show the expected security-cost trade-off. Higher admissibility thresholds reduce false admission but increase false rejection and operating cost. Higher attack intensity increases cyber exposure and cost because the optimizer must rely more heavily on restricted admissibility, conservative dispatch, and fallback-oriented safeguards [22]. High uncertainty pressure also increases cost because feasible-domain contraction becomes more restrictive when renewable, load, price, and flexibility uncertainty intensify.

4.11. Computational Performance

Table 13 reports the runtime decomposition of the proposed workflow. The optimization solver accounts for the largest share of computation time, while trust scoring, contamination estimation, admissibility screening, and audit-record construction remain lightweight. The total runtime on the modified IEEE 118-bus case is 2.64 s, which remains below typical real-time market-clearing intervals.

The runtime results indicate that the proposed workflow is computationally practical for benchmark-scale real-time clearing. Even for the IEEE 300-bus case, the total runtime remains below 10 s in the representative configuration. This supports the claim that the agentic verification layer does not dominate computational cost; instead, the main runtime driver remains the network-constrained clearing optimization.

Table 12: Sensitivity of the proposed method to key parameters.

Parameter setting	Cyber exposure	False admission	False rejection	Cost increase
Low admissibility threshold	0.241	8.2%	3.9%	3.71%
Baseline admissibility threshold	0.183	4.6%	6.2%	5.16%
High admissibility threshold	0.151	2.9%	11.4%	7.83%
Low attack intensity	0.112	2.7%	4.9%	3.94%
Medium attack intensity	0.183	4.6%	6.2%	5.16%
High attack intensity	0.294	7.5%	8.7%	8.61%
High uncertainty pressure	0.231	5.9%	9.3%	9.24%

Table 13: Runtime decomposition of the proposed zero-trust clearing workflow.

Workflow component	IEEE 30-bus	IEEE 118-bus	IEEE 300-bus
Trust scoring (s)	0.07	0.16	0.38
Contamination estimation (s)	0.05	0.12	0.31
Admissibility screening (s)	0.04	0.09	0.22
Optimization solving (s)	0.61	1.84	5.73
Audit-record construction (s)	0.06	0.18	0.47
Total runtime (s)	0.83	2.64	7.11

4.12. Summary of Validated Findings

The benchmark validation supports four main findings. First, the proposed method improves cyber-resilient admissibility by reducing false admission while avoiding the excessive rejection behavior of rigid rule-based zero trust. Second, it improves operational feasibility under cyber-contaminated conditions by reducing line-limit violations, price-bound violations, and load shedding. Third, it strengthens settlement legitimacy by preserving audit evidence and penalizing weak settlement defensibility within the clearing objective. Fourth, it maintains computational practicality because the additional agentic verification and auditability steps remain lightweight relative to the optimization problem.

These findings should be interpreted within the limits of benchmark-based simulation. The results support the methodological validity of the proposed framework under controlled cyber-attack and uncertainty scenarios, but they do not yet establish deployment-level performance on utility-grade cyber-physical logs, real market operator data, or adversarial digital-twin environments.

5. Discussion and Conclusion

5.1. Discussion

The proposed zero-trust agentic AI framework reframes cyber-resilient energy market clearing as an integrated problem of verification, admissibility, optimization, governance, and auditability. Unlike conventional clearing models that assume the reliability of bids, measurements, dispatch signals, flexibility responses, and settlement records, the proposed approach treats every market input as conditionally admissible. This design is especially important for real-time balancing markets, where cyber-contaminated information can directly affect dispatch feasibility, price formation, flexibility activation, and settlement legitimacy.

The main methodological contribution is the integration of agentic AI into the market-clearing process without allowing autonomous agents to act as unconstrained market operators. The agentic layer evaluates evidence, estimates trust, detects residual contamination, and supports admissibility decisions, while the optimization layer remains responsible for feasible dispatch and pricing. This separation is important because it preserves the benefits of autonomous reasoning while maintaining mathematical control, operational transparency, and human-governed intervention for high-impact actions.

The results indicate that the proposed framework can reduce residual cyber-risk exposure and false admission while preserving market feasibility. Compared with conventional clearing, the proposed model introduces moderate operational cost increases, but these costs are justified by improved cyber resilience, lower load shedding, and stronger settlement legitimacy. Compared with rigid rule-based zero-trust screening, the proposed model avoids excessive conservatism by using calibrated trust, agentic consensus, residual cyber contamination, and uncertainty pressure rather than relying on fixed rejection rules.

A further strength of the framework is its explicit treatment of settlement legitimacy. In cyber-degraded market conditions, technically feasible dispatch is not sufficient if the underlying evidence cannot support defensible settlement. By modeling bid legitimacy, measurement legitimacy, dispatch-execution legitimacy, flexibility-response legitimacy, and audit-log legitimacy, the proposed approach connects real-time clearing decisions with post-market accountability. This is critical for practical market operation because cyber incidents may create disputes not only over dispatch outcomes but also over financial settlement and regulatory compliance.

The proposed return-to-normal rule also contributes to operational robustness. Instead of restoring normal clearing after a single improvement in cyber indicators, the model requires sustained recovery across trust, residual contamination, settlement legitimacy, and normal-domain feasibility. This prevents premature restoration and supports a controlled transition from zero-trust cyber-resilient clearing back to standard market operation.

5.2. Practical Implications

The proposed framework is suitable for energy market operators, transmission system operators, distribution system operators, aggregators, and cyber-security operation centers responsible for critical energy infrastructure. It can support real-time decision-making during cyber-degraded conditions by identifying which market channels remain admissible, which inputs require rejection or escalation, and when fallback intervention is necessary. The framework also provides an auditable structure for explaining why certain bids, measurements, flexibility responses, or settlement records were admitted or rejected.

For implementation, the model can be deployed as a supervisory decision-support layer above existing market-clearing engines. The agentic AI layer does not need to replace the market optimizer; instead, it can provide trust scores, contamination indicators, admissibility flags, and audit records that modify the feasible domain and objective function of the existing clearing process. This makes the framework more realistic for gradual adoption in operational environments.

5.3. Limitations

The proposed study has several limitations. First, the evaluation is designed around benchmark power-system test cases and simulation-based cyberattack scenarios; therefore, real utility data would be required to validate the framework under operational deployment conditions. Second, the performance of the agentic verification layer depends on the quality of evidence sources, cyber logs, historical baselines, and anomaly indicators. If these inputs are incomplete or manipulated, trust estimation may become less reliable. Third, the proposed formulation assumes bounded attacker capability and does not model an adversary capable of fully compromising all agents, measurements, and audit records simultaneously.

Another limitation is computational scalability under very large-scale systems. Although the proposed structure is suitable for real-time intervals such as 5-minute or 15-minute balancing markets, larger networks, higher-resolution uncertainty modeling, and complex mixed-integer formulations may increase solve time. Future implementation should therefore examine decomposition, parallel agent execution, and fast convex approximations.

5.4. Future Research

Future research should implement the proposed model on larger benchmark systems, such as IEEE 300-bus and synthetic transmission networks, and compare its performance under different attack intensities, renewable penetration levels, and flexibility availability. Another important direction is to develop adaptive learning mechanisms for updating agent weights, evidence reliability, and admissibility thresholds based on post-clearing audit outcomes. Future work should also examine adversarial attacks against the agentic AI layer itself, including prompt injection, model manipulation, coordinated false evidence, and compromised audit logs.

A further extension is to integrate the proposed framework with digital twins of cyber-physical energy systems. This would allow the agentic layer to test whether suspicious cyber evidence is consistent with physical grid behavior before the clearing decision is finalized. Finally, regulatory and market-design implications should be studied, especially regarding settlement disputes, operator liability, transparency requirements, and human approval rules during cyber-degraded market operation.

5.5. Conclusion

This paper proposed a zero-trust agentic AI framework for cyber-resilient energy market clearing. The framework integrates multi-agent verification, trust scoring, residual cyber-contamination modeling, uncertainty-pressure assessment, admissibility screening, feasible-domain contraction, and auditable market optimization. The central idea is that real-time market inputs should not be trusted by default during cyber-degraded operation; instead, they should be admitted into the clearing process only after agentic verification and zero-trust screening.

The proposed model advances existing cyber-aware market-clearing approaches by linking cybersecurity evidence directly to optimization decisions. It enables the market to continue operating under degraded digital trust while reducing cyber exposure, limiting false admission, preserving physical feasibility, and improving settlement legitimacy. The framework also maintains human authority over high-impact interventions and return-to-normal certification, making it more suitable for critical infrastructure contexts.

Overall, the study demonstrates that cyber-resilient energy market clearing should not be treated only as an anomaly-detection problem or only as a robust optimization problem. It should be modeled as a governed cyber-physical market process in which agentic AI, zero-trust verification, admissibility control, optimization, and auditability operate together. This integrated perspective provides a methodological foundation for auditable, human-governed, and cyber-resilient energy market operation under residual cyber contamination and dynamic uncertainty.

Declarations

Ethics Approval and Consent to Participate

Not applicable. This study does not involve human participants, human data, animal subjects, or clinical materials.

Consent for Publication

Not applicable.

Availability of Data and Materials

The benchmark validation is based on standard IEEE test-system structures and the parameterized simulation settings described in the manuscript. The generated simulation outputs, parameter settings, and implementation materials used to support the reported results are available from the corresponding author upon reasonable request.

Competing Interests

The authors declare that they have no competing interests.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Authors' Contributions

S.M. contributed to the conceptualization, methodological design, mathematical formulation, and manuscript preparation. M.R.S. contributed to the modeling framework, validation design, result interpretation, and manuscript revision. Both authors reviewed and approved the final version of the manuscript.

Acknowledgements

The authors would like to acknowledge the use of standard IEEE benchmark system structures, which supported the methodological validation of the proposed framework.

Use of AI-Assisted Tools

AI-assisted tools were used only for language refinement, formatting support, and editorial improvement. The scientific content, mathematical formulation, methodological design, interpretation of results, and final responsibility for the manuscript remain entirely with the authors.

References

[1] S. Rose, O. Borchert, S. Mitchell, S. Connelly, Zero trust architecture, Special Publication 800-207, National Institute of Standards and Technology (2020). doi:10.6028/NIST.SP.800-207.

[2] E. Tabassi, Artificial intelligence risk management framework (AI RMF 1.0), NIST AI 100-1, National Institute of Standards and Technology (2023). doi:10.6028/NIST.AI.100-1.

[3] R. Sepehrzad, H. Hosseinalibeiki, N. Taghinezhad, N. Khosravi, A. Al Durra, M. S. Sadabadi, Enhanced cyber-resilience in flexible energy markets for microgrids: A trust-aware federated deep reinforcement learning framework, Applied Energy 406 (2026) 127282. doi:10.1016/j.apenergy.2025.127282.

[4] Y. Liu, P. Ning, M. K. Reiter, False data injection attacks against state estimation in electric power grids, ACM Transactions on Information and System Security 14 (1) (2011) 13. doi:10.1145/1952982.1952995.

[5] R. Sepehrzad, H. Hosseinalibeiki, M. S. Sadabadi, A. Al Durra, Microgrid secure pricing and real-time flexible energy market integrated with trust-based demand response: A multi-layer fdrl architecture, International Journal of Electrical Power & Energy Systems 174 (2026) 111536. doi:10.1016/j.ijepes.2025.111536.

[6] A. S. George, Architectural convergence in security operations: A technical framework for ai-augmented threat detection, automated response, and organizational cyber resilience, Partners Universal Innovative Research Publication 4 (2) (2026) 67–91.

[7] M. A. Islam, S. Somu, F. M. F. Aldaihani, The rise of agentic ai: Synthesis of current knowledge and future research agenda, Global Business and Organizational Excellence 45 (3) (2026) 402–416.

[8] H. Hosseinalibeiki, M. Heidari, A conceptual hybrid model based on ethereum for implementing sustainable development goals, European Journal of Sustainable Development Research 8 (4) (2024) em0272. doi:10.29333/ejosdr/15434.

[9] A. M. Almahdi, M. I. Mohamed, Blockchain-cim governance framework for smart cities: Simulation, prototype, and cross-regional validation, African Journal of Advanced Pure and Applied Sciences (2026) 491–512.

[10] L. Xie, Y. Mo, B. Sinopoli, Integrity data attacks in power market operations, IEEE Transactions on Smart Grid 2 (4) (2011) 659–666. doi:10.1109/TSG.2011.2161892.

[11] R. Sepehrzad, A. Kazemian, N. Khosravi, J. Watson, H. Hosseinalibeiki, A. Al Durra, M. S. Sadabadi, Risk-aware multi-agent reinforcement learning and blockchain enforcement for peer-to-peer trading market and demand response in residential energy communities, SSRN (2026). doi:10.2139/ssrn.6559454.
URL `https://ssrn.com/abstract=6559454`

[12] A. Dzafic, O. Elma, U. Bayram, A comprehensive review of artificial intelligence methods in energy system applications, International Journal of Electrical Power & Energy Systems 177 (2026) 111667. doi:10.1016/j.ijepes.2026.111667.

[13] Y. Mo, T. H.-J. Kim, K. Brancik, D. Dickinson, H. Lee, A. Perrig, B. Sinopoli, Cyber-physical security of a smart grid infrastructure, Proceedings of the IEEE 100 (1) (2012) 195–209. doi:10.1109/JPROC.2011.2161428.

[14] R. Sepehrzad, M. Moafi, N. Khosravi, H. Hosseinalibeiki, A. Al Durra, M. S. Sadabadi, Retail-informed secure wholesale energy market and bid generation with projection clearing exchange for networked microgrids: A large language model-assisted multi-agent deep reinforcement learning approach, SSRN (2026). doi:10.2139/ssrn.5991242.

[15] S. Cheng, Y. Yi, T. Liu, Optimal defense strategy for a platform supply chain considering uncertainty cyberattacks under blockchain technology, Expert Systems with Applications 318 (2026) 132091. doi:10.1016/j.eswa.2026.132091.

[16] R. Sepehrzad, N. Khosravi, M. Moafi, H. Hosseinalibeiki, A. Al Durra, M. S. Sadabadi, A safe, secure, and uncertainty-aware demand response architecture using llm-guided hierarchical multi-agent deep reinforcement learning in integrated energy system, SSRN (2026). doi:10.2139/ssrn.6025576.

[17] R. Sepehrzad, N. Khosravi, M. Moafi, H. Hosseinalibeiki, A. Al Durra, M. S. Sadabadi, Risk-aware secure microgrid pricing mechanism validated with hybrid llm-blockchain considering hybrid energy storage system and demand response flexibility under uncertainty and cyberattacks, SSRN (2026). doi:10.2139/ssrn.5956836.

[18] A. Gaeta, H. Hosseinalibeiki, V. Loia, F. Orciuoli, Real-time trust-weighted consensus and auditing of llms for secure propaganda detection, Applied Soft Computing 201 (2026) 115459. doi:10.1016/j.asoc.2026.115459.
URL https://www.sciencedirect.com/science/article/pii/S1568494626009075

[19] S. Tan, D. De, W. Song, J. Yang, S. K. Das, Survey of security advances in smart grid: A data driven approach, IEEE Communications Surveys & Tutorials 19 (1) (2017) 397–422. doi:10.1109/COMST.2016.2616442.

[20] H. He, J. Yan, Cyber-physical attacks and defences in the smart grid: A survey, IET Cyber-Physical Systems: Theory & Applications 1 (1) (2016) 13–27. doi:10.1049/iet-cps.2016.0019.

[21] R. Sepehrzad, H. Hosseinalibeiki, N. Khosravi, A. Al Durra, Post-disruption resilient recovery of microgrid real-time balancing markets under dynamic uncertainty and market-state degradation, SSRN (2026). doi:10.2139/ssrn.6727505.

[22] R. Sepehrzad, H. Hosseinalibeiki, N. Khosravi, A. Al Durra, Resilience-oriented market-to-delivery demand response in integrated energy systems under dynamic uncertainty and cyber-physical disruptions: A constrained deep reinforcement learning framework, SSRN (2026). doi:10.2139/ssrn.6727499.

[23] O. A. Mohammed, M. H. Cintuglu, A. S. Uluagac, K. Akkaya, A survey on smart grid cyber-physical system testbeds, IEEE Communications Surveys & Tutorials 19 (1) (2017) 446–464. doi:10.1109/COMST.2016.2627399.

[24] R. Sepehrzad, H. Hosseinalibeiki, H. Vatankhah Ghadim, N. Khosravi, A. Al Durra, M. S. Sadabadi, Energy storage system-based demand response with time-varying uncertainties, cyber-degraded observability, and risk-aware settlement in renewable-rich integrated energy systems, SSRN (2026). doi:10.2139/ssrn.6668103.

[25] A. Giani, E. Bitar, M. Garcia, M. McQueen, P. Khargonekar, K. Poolla, Smart grid data integrity attacks, IEEE Transactions on Smart Grid 4 (3) (2013) 1244–1253. doi:10.1109/TSG.2013.2245155.

[26] R. Sepehrzad, H. Hosseinalibeiki, N. Khosravi, A. Al Durra, Credibility-aware demand response continuation and regime-switching market operation in multi-energy microgrids under hybrid cyber threats and dynamic uncertainty, SSRN (2026). doi:10.2139/ssrn.6667538.

[27] R. Sepehrzad, H. Hosseinalibeiki, N. Khosravi, A. Al Durra, M. S. Sadabadi, Heterogeneous energy storage-supported demand response in integrated energy systems: A resilient market-to-delivery framework under compound disturbances, SSRN (2026). doi:10.2139/ssrn.6811358.
URL https://ssrn.com/abstract=6811358

[28] Y. Xu, A review of cyber security risks of power systems: From static to dynamic false data attacks, Protection and Control of Modern Power Systems 5 (1) (2020) 19. doi:10.1186/s41601-020-00164-w.

[29] D. Bertsimas, E. Litvinov, X. A. Sun, J. Zhao, T. Zheng, Adaptive robust optimization for the security constrained unit commitment problem, IEEE Transactions on Power Systems 28 (1) (2013) 52–63. doi:10.1109/TPWRS.2012.2205021.

[30] A. Lorca, X. A. Sun, E. Litvinov, T. Zheng, Multistage adaptive robust optimization for the unit commitment problem, Operations Research 64 (1) (2016) 32–51. doi:10.1287/opre.2015.1456.

[31] R. D. Zimmerman, C. E. Murillo-Sánchez, R. J. Thomas, Matpower: Steady-state operations, planning, and analysis tools for power systems research and education, IEEE Transactions on Power Systems 26 (1) (2011) 12–19. doi:10.1109/TPWRS.2010.2051168.

[32] H. Badrsimaei, R.-A. Hooshmand, S. Nobakhtian, Monte-carlo-based data injection attack on electricity markets with network parametric and topology uncertainties, International Journal of Electrical Power & Energy Systems 138 (2022) 107915. doi:10.1016/j.ijepes.2021.107915.

[33] G. O. Alsharif, C. Anagnostopoulos, A. K. Marnerides, Energy market manipulation via false-data injection attacks: A review, IEEE Access 13 (2025) 42559–42573. doi:10.1109/ACCESS.2025.3548914.

[34] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, I. Mordatch, Multi-agent actor-critic for mixed cooperative-competitive environments, in: Advances in Neural Information Processing Systems, Vol. 30, 2017.

[35] A. Gholami, K. Sun, S. Zhang, X. A. Sun, An admm-based distributed optimization method for solving security-constrained alternating current optimal power flow, Operations Research 71 (6) (2023) 2045–2060. doi:10.1287/opre.2023.2486.

[36] R. Louca, E. Bitar, Robust ac optimal power flow, IEEE Transactions on Power Systems 34 (3) (2019) 1669–1681. doi:10.1109/TPWRS.2018.2849581.

[37] D. Bienstock, M. Chertkov, S. Harnett, Chance-constrained optimal power flow: Risk-aware network control under uncertainty, SIAM Review 56 (3) (2014) 461–495. doi:10.1137/130910312.

[38] A. Teixeira, K. C. Sou, H. Sandberg, K. H. Johansson, Secure control systems: A quantitative risk management approach, IEEE Control Systems Magazine 35 (1) (2015) 24–45. doi:10.1109/MCS.2014.2364709.

[39] J. Zhang, Z. Chu, L. Sankar, O. Kosut, Can attackers with limited information exploit historical data to mount successful false data injection attacks on power systems?, IEEE Transactions on Power Systems 33 (5) (2018) 4775–4786. doi:10.1109/TPWRS.2018.2818746.

[40] S. Lakshminarayana, A. Kammoun, M. Debbah, H. V. Poor, Data-driven false data injection attacks against power grids: A random matrix approach, IEEE Transactions on Smart Grid 12 (1) (2021) 635–646. doi:10.1109/TSG.2020.3011391.

[41] J. Liang, L. Sankar, O. Kosut, Vulnerability analysis and consequences of false data injection attack on power system state estimation, IEEE Transactions on Power Systems 31 (5) (2016) 3864–3872. doi:10.1109/TPWRS.2015.2504950.

[42] H. T. Reda, A. Anwar, A. Mahmood, Comprehensive survey and taxonomies of false data injection attacks in smart grids: Attack models, targets, and impacts, Renewable and Sustainable Energy Reviews 163 (2022) 112423. doi:10.1016/j.rser.2022.112423.

[43] S. Aoufi, A. Derhab, M. Guerroumi, Survey of false data injection in smart power grid: Attacks, countermeasures and challenges, Journal of Information Security and Applications 54 (2020) 102518. doi:10.1016/j.jisa.2020.102518.

[44] V. Y. Pillitteri, T. L. Brewer, Guidelines for smart grid cybersecurity, NIST Interagency/Internal Report 7628 Revision 1, National Institute of Standards and Technology (2014). doi:10.6028/NIST.IR.7628r1.

[45] O. Kosut, L. Jia, R. J. Thomas, L. Tong, Malicious data attacks on smart grid state estimation: Attack strategies and countermeasures, in: 2010 First IEEE International Conference on Smart Grid Communications, 2010, pp. 220–225. doi:10.1109/SMARTGRID.2010.5622045.

[46] G. Hug, J. A. Giampapa, Vulnerability assessment of ac state estimation with respect to false data injection cyber-attacks, IEEE Transactions on Smart Grid 3 (3) (2012) 1362–1370. doi:10.1109/TSG.2012.2195338.

[47] Y. He, G. J. Mendis, J. Wei, Real-time detection of false data injection attacks in smart grid: A deep learning-based intelligent mechanism, IEEE Transactions on Smart Grid 8 (5) (2017) 2505–2516. doi:10.1109/TSG.2017.2703842.

[48] A. S. Musleh, G. Chen, Z. Dong, A survey on the detection algorithms for false data injection attacks in smart grids, IEEE Transactions on Smart Grid 11 (3) (2020) 2218–2234. doi:10.1109/TSG.2019.2949998.